

San Francisco Bay Regional Water Quality Control Board

July 1, 2022
CIWQS Place No. 236597

Las Gallinas Valley Sanitary District
Attention: Chris DeGabriele, Interim General Manager
300 Smith Ranch Road
San Rafael, CA 94903
Sent electronically to cdegabriele@lqvsvd.org

Subject: Las Gallinas Valley Sanitary District, Marin County – Enrollment under General Waste Discharge Requirements for the Discharge of Biosolids to Land for Use as a Soil Amendment in Agricultural Activities, Order No. 2004-0012-DWQ

Dear Mr. DeGabriele:

This letter enrolls Las Gallinas Valley Sanitary District (Discharger) under the General Waste Discharge Requirements for the Discharge of Biosolids to Land for Use as a Soil Amendment in Agricultural, Silvicultural, Horticultural, and Land Reclamation Activities, Order No. 2004-0012-DWQ (General Order). This enrollment conditionally authorizes the land application of biosolids, generated from the Discharger's wastewater treatment plant at 300 Smith Ranch Road, San Rafael, in Marin County (Facility), to agricultural areas on District property.

On March 9, 2021, the Discharger submitted a Notice of Intent (NOI) to the San Francisco Bay Regional Water Quality Control Board (Water Board) for coverage under the General Order. Supplemental information was submitted on July 28, 2021. Based on the information presented in the NOI and subsequent documents, the land application of biosolids as described (the Project) is consistent with the General Order's requirements.

The Water Board hereby finds that the General Order is applicable to the Project and that the Discharger is henceforth enrolled under and regulated by the General Order, provided that the Discharger complies with the conditions contained in this letter.

This letter serves as the Notice of Applicability (NOA) for enrollment under the General Order. Due to the Project site conditions, including proximity to the San Francisco Bay and tidal wetlands, shallow groundwater, and surface water drainage, Water Board staff will review this NOA after at least three years of data is available to assess whether the General Order continues to be the appropriate long-term permitting mechanism for the

Project. A site-specific Monitoring and Reporting Program for the Project is included as NOA Att. 1.

The discharge must be managed in accordance with the requirements contained in the General Order, the information submitted in the NOI, and the requirements contained in this NOA. A copy of the General Order is enclosed as Att. 2; it may also be viewed on the Water Board website.

(https://www.waterboards.ca.gov/board_decisions/adopted_orders/water_quality/2004/wqo/wqo2004-0012.pdf)

I. PROJECT LOCATION

District property where biosolids will be applied is located northeast of the Facility and include Assessor's Parcel Numbers (APNs) 155-011-13, 155-011-14, and 155-011-33, as shown in the site plan (Att. 3). These parcels are adjacent to San Pablo Bay to the east and primarily open space to the south and west. This land formerly consisted of mudflats and marshlands that have been separated from tidal action by constructing levees along the east and south boundaries in the early 1900s.

The closest residences are located approximately 0.3 miles to the west of parcel 155-01-13. The land directly adjacent to parcel 155-011-33 on the west side is an unused land owned by the Silveira San Rafael Ranch and the land to the north is an unused land previously owned by the St. Vincent's School for Boys. McInnis County Park and Santa Venetia Marsh are located south, approximately 0.5 and 1 miles away, respectively, and the Hamilton Wetlands are located due north.

The geographic location of the property and the biosolids application areas described in this NOA are within the jurisdiction of the Water Board and subject to the San Francisco Bay Basin Water Quality Control Plan (Basin Plan). The Basin Plan designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies for protecting waters of the basin and incorporates by reference State Water Board plans and policies. This NOA and its requirements apply to all biosolids land application areas described herein.

II. PROJECT DESCRIPTION

The Discharger has proposed to land apply Class B biosolids for agricultural use on up to 304 acres of District property (parcels 155-011-13, 155-011-14, and 155-011-33, shown on the site plan in Att. 3). This Project will transition the District from biosolids disposal to biosolids beneficial use.

Biosolids generated from the Facility are treated by gravity thickening and anaerobic digestion in primary and secondary digesters and then pumped to three sludge storage lagoons. The sludge storage lagoons are double-lined and have a total capacity of approximately 3.2 million gallons. Currently, between 200 and 250 dry metric tons of biosolids per year are disposed of onsite at the Discharger's nine-acre dedicated land disposal (DLD) site, shown on the site plan (Att. 3). The Discharger injects liquid biosolids from the sludge storage ponds to the DLD site one to two times per year.

Through this Project, the Discharger will shift from disposing biosolids on nine acres to beneficially using biosolids by land applying on up to 304 acres as a soil amendment for agricultural production. Parcels 155-011-13 and 155-011-14 (238 acres) are currently under agricultural production (pasture hay crop) and irrigated with treated effluent (recycled or reclaimed water). Parcel 155-011-33 (66 acres) is currently unused land that has been disked in the past for weed abatement and is not irrigated; it is planned to be dry farmed for pasture hay crop. Liquid or cake biosolids will be land applied at agronomic rates. In the liquid form (approximately 4% solids), biosolids will be applied using either mobile applicator vehicles and equipment or direct pumping from three existing sludge storage lagoons to the biosolids application areas. Dewatered biosolids cake application may be done at some time in the future at the discretion of the District, at percent solids estimated between 11 and 20%, using a mobile belt press, centrifuge system, or equal, operated by the District or a third-party contractor to the District. A third-party contractor or contractors will be used for biosolids land application and farming services. No in-field storage of liquid biosolids is planned; management of dewatered biosolids cake will comply with General Order requirements. The DLD site will remain in place and be available to dispose biosolids if needed.

The Discharger plans to apply 200 dry metric tons of biosolids annually. Land application is planned to occur between May 1 and October 31 on parcel 155-011-33 and between May 1 and August 15 on parcels 155-011-13 and 155-011-14 to reduce runoff risk and allow crop uptake of nutrients throughout the growing season before harvest.

III. SITE CHARACTERISTICS

The site is located in the Bay Plain geomorphic zone within the Coast Ranges Geomorphic Province, a seismically active province that is characterized by a series of northwest trending faults, mountain ranges, and valleys. The Bay Plain extends from the edge of San Pablo Bay to the foot of the hills west of the site (Jones & Stokes Associates, Inc., 1998). The soils at the site are classified as Reyes series, which occur on estuarine sediments in "reclaimed" diked tidelands and are formed in mixed alluvium from the Bay and streams (National Cooperative Soil Survey, 1985). Underlying the Reyes soil are thick deposits of soft, unconsolidated, organic-rich, highly plastic, water-saturated, silty clays known as Bay Mud. The depth of Bay Mud in the area ranges from 30 to 90 feet below the ground surface (Jones & Stokes Associates, Inc., 1998). Bay Mud is typically classified as an unconfined aquitard due to its low permeability and the absence of a confining layer (The Earth Technology Corporation, 1994).

A. Surface Water and Site Drainage

The topographic maps (USGS, 2018a, 2018b) and aerial photographs (Google Earth, 2020) show sloughs and drainage channels present at and surrounding the site. The general topography at the site is flat with a slight slope to the east with elevations ranging between 4 and -1 feet above mean sea level (MSL). The expected average annual precipitation in the area is 34 inches. The Discharger operates a pumphouse by the levee access road at the southeastern corner of parcel 155-011-14, which is used for runoff control and flood management when needed.

According to historical topographic maps, Miller Creek previously flowed from the west to San Pablo Bay along the north border of parcel 155-011-33 until it was rerouted in the 1940s to flow south along the parcel's west border and then east along the south border of the LGVSD facility (USGS, 1914 and U.S. Army Corps of Engineers, 1942).

B. Groundwater

Groundwater in the vicinity of site is not used as drinking water or for agriculture. The depth to groundwater ranges from near zero to 8 feet below ground surface west of the levees, which is within Bay Mud. Water levels in the monitoring wells at sites in the area remain relatively static and do not indicate a hydraulic gradient from a sloping water table. The hydraulic gradient at nearby sites appears to be tidally influenced and flow perpendicular to the San Pablo Bay edge or toward drainage channels and creeks that exhibit tidal flows (The Earth Technology Corporation, 1994). Groundwater flow direction at parcel 155-011-33 may be influenced by the present and former routes of Miller Creek. Based on the tidal influence exhibited at nearby sites, flat topography, low elevation, and the surface water flow in the area, groundwater flow at the site is anticipated to be tidally influenced and flow west to east and east to west.

IV. CONDITIONS OF ENROLLMENT IN GENERAL ORDER

The Water Board may, in accordance with the General Order, require modifications to proposed discharge activities or additional information to verify that the proposed discharge will not cause or contribute to violations of water quality standards or otherwise impact water quality. Due to the proximity to the Bay and tidal marshland, shallow groundwater, and potential for surface water impacts from runoff and flooding on the parcels, additional conditions for this Project are necessary. Accordingly, in order to maintain coverage under the General Order, the Discharger is required to:

- Participate in the research project *Unregulated Organic Chemicals in Biosolids: Prioritization, Fate and Risk Evaluation for Land Applications*, EPA Grant Number R840245 (Research Project). Coordinate with the Water Board as appropriate, provide timely updates on research progress, and provide project status reports in each annual report for the duration of the Research Project.
- Conduct a biological site assessment of the land application sites prior to land application. In addition to the biological site assessment requirements described in the General Order, Pre-Application Report No. 11, the assessment shall evaluate the potential presence and extent of wetlands. This NOA does not authorize the land application of biosolids to any seasonal wetlands or other waters of the United States.¹

¹ Code of Federal Regulations, title 40, part 503.14(b) (40 CFR 503.14(b)) states: "Bulk sewage sludge shall not be applied to agricultural land, forest, a public contact site, or a reclamation site that is flooded, frozen, or snow-covered so that the bulk sewage sludge enters a wetland or other waters of the United States, as defined in 40 CFR 122.2, except as provided in a permit issued pursuant to section 402 or 404 of the CWA."

- Complete additional characterization of Discharger biosolids and biosolids disposal and land application sites, as specified in the required technical reports: Biosolids and Site Characterization Workplan and Report.
- Provide a flood protection plan that assesses the vulnerability of the Project to sea level rise and groundwater rise, as specified in the required technical report: Flood Protection Plan.
- Limit the land application period to between May 1 and October 31 on parcel 155-011-33 and between May 1 and August 15 on parcels 155-011-13 and 155-011-14 to reduce runoff risk.
- Conduct additional monitoring as specified in the site-specific Monitoring and Reporting Program.

V. SITE SPECIFIC AND OTHER REQUIREMENTS

- Application of biosolids at a location or in a manner different from that as described in the NOI, this NOA, and the Pre-Application Reports is prohibited.
- The application shall not cause or threaten to cause pollution as defined by California Water Code (Water Code) section 13050.
- There shall be no discharge of biosolids from the storage or application areas to adjacent land areas not regulated by this NOA, to surface water, or to surface water drainage courses.
- The application of biosolids shall comply with all applicable setbacks contained in the General Order.
- Application of biosolids at rates in excess of the nitrogen requirements of the vegetation or at rates that would degrade groundwater quality is prohibited.
- The application of biosolids to water-saturated ground or during periods of precipitation is prohibited.
- The application of Class B biosolids containing a moisture content of less than 50 percent is prohibited.
- The Discharger is required to implement its Monitoring and Reporting Plan.

VI. MONITORING AND REPORTING PROGRAM

The Discharger shall comply with the site-specific Monitoring and Reporting Program in Notice Att. 1, which is attached hereto and made part of this Notice by reference.

VII. TECHNICAL REPORT REQUIREMENTS

The following technical reports are required pursuant to Water Code sections 13383 and 13267. All technical documents shall be signed by and stamped with the seal of a California registered geologist, a California certified engineering geologist, or a California registered civil engineer. The technical reports shall be acceptable to the Water Board's Executive Officer and shall be submitted as described below.

A. Biosolids and Site Characterization Workplan

By February 1, 2023, the Discharger shall submit a workplan to investigate the occurrence of constituents in the Facility biosolids and in soil in and adjacent to the area where biosolids are disposed of and will be applied to the land, as well as groundwater and surface waters receiving stormwater runoff from areas where biosolids are applied to land. Characterization of baseline conditions and conditions after land application occurs shall be included in the workplan; existing historical data may be used to inform baseline conditions. It is anticipated that the District will coordinate with the Research Project in development of this workplan.

At a minimum, the workplan shall include the following:

- Site description and historical property use
- A description of the geology, hydrogeology, and surface water hydrology of the property and vicinity
- Summary of existing information and soil or water quality data, if any, at the property and vicinity, including an evaluation of their relevance to current conditions
- A map showing the proposed sampling locations by media (e.g., soil, groundwater, surface water) with a table or text presenting the rationale for each location
- A description of field sampling methods to be used
- A list of target constituents for laboratory analysis by media (e.g., biosolids, soil, groundwater, and surface water) for a one-time sampling and analysis, rationale for constituent selection, and identification of laboratory analytical methods to be used²; the list of target constituents to consider³:
 - Constituents required by the General Order
 - EPA Priority Pollutant list (not otherwise included by the General Order)
 - Polycyclic Aromatic Hydrocarbons (PAHs)
 - Volatile Organic Compounds (VOCs)
 - Total Petroleum Hydrocarbon (TPH)
 - Dioxins and furans
 - Per- and Polyfluoroalkyl Substances (PFAS)
 - Flame retardants such as polybrominated diphenyl ethers (PBDEs)

² Laboratory reporting limits shall be lower than concentrations that implement applicable limits, water quality objectives, or risk thresholds for the constituents to be analyzed, as reasonable.

³ The Discharger may propose other constituents or to omit suggested constituents. The list can vary between environmental media, and the Discharger may propose to analyze the biosolids for a more comprehensive list which informs a more targeted list to analyze in other environmental matrices.

- Relevant endocrine disruptors such as bisphenols, phthalates, and triclosan
 - Ethoxylated surfactants
 - Pesticides including pyrethroids, imidacloprid, and fipronil
- A basis for evaluating the soil, groundwater, and surface water data with suitable ambient background conditions, water quality objectives, or other screening guidelines for the target constituents
 - Biosolids and field soils data shall be evaluated against ambient background conditions; the General Order; *Standards for the Use or Disposal of Sewage Sludge*, Code of Federal Regulations, title 40, part 503 (40 CFR Part 503); *Beneficial Reuse of Dredged Materials: Sediment Screening and Testing Guidelines*, San Francisco Bay Regional Water Quality Control Board Draft Staff Report (May 2000) and *Expert review of the sediment screening guidelines for the beneficial reuse of dredged material in San Francisco Bay*⁴ (2020); San Francisco Bay Regional Water Quality Control Board Environmental Screening Levels (ESLs)⁵; and peer-reviewed toxicity thresholds for sediment (for constituents of emerging concern not included in the prior references).
 - Groundwater and surface water data shall be evaluated against ambient background conditions; the General Order; 40 CFR Part 503; ESLs; and peer-reviewed toxicity thresholds for freshwater and marine water.
 - Exceedances of thresholds or screening levels which are not regulatory limits are not considered a violation of this NOA.
- Identification of permits or authorizations to be obtained.
- A schedule for the field implementation of the workplan, data evaluation, and reporting of results.

B. Biosolids and Site Characterization Report

The Discharger shall submit a technical report documenting implementation of the workplan as approved by the Water Board. At a minimum, the report shall include the following:

- Description of preparatory and investigation activities
- Description of the laboratory analytical results for biosolids, soil, groundwater, and surface water
- Data evaluation
- Conclusions and recommendations

⁴ Foley, M.; Christian, E.; Goeden, B.; Ross, B. 2020. Expert review of the sediment screening guidelines for the beneficial reuse of dredged material in San Francisco Bay. SFEI Contribution No. 978. San Francisco Estuary Institute: Richmond, CA.

⁵ https://www.waterboards.ca.gov/sanfranciscobay/water_issues/programs/esl.html

- A map or maps showing the final locations of field sampling locations and depicting analytical results, as appropriate
- Tables of analytical results
- Copies of laboratory analytical reports and other information (e.g., boring logs, well construction logs)

C. Sampling and Analysis Plan

By February 1, 2023, the Discharger shall submit a sampling and analysis plan sufficient to assure compliance with the terms of the General Order, NOA, and Monitoring and Reporting Program. At a minimum, the sampling and analysis plan shall describe the following:

- Sample chain-of-custody procedures and documentation
- Sampling locations
- Sampling frequencies
- Sample handling/preservation procedures
- Analytical methods
- Sample containers, preservatives, and holding times

D. Flood Protection Plan

By February 1, 2023, the Discharger shall submit a Flood Protection Plan that includes an assessment of the vulnerability of the Project to sea level rise and groundwater rise, as follows:

- **Sea Level Rise.** Explain how the Discharger manages existing flooding risks at the Project location (e.g., protective measures already in place, planned, or proposed). Explain how the Discharger intends to manage future flooding risks through the expected life of the Project or through 2050 (e.g., planning efforts and protective measures in place, planned, or proposed). What guidance and assumptions are being used to anticipate sea level rise? If the Discharger has not yet established a plan, explain the process and timeline for doing so.
- **Groundwater Rise.** Explain how the Discharger intends to manage future flooding at the Project location related to groundwater rise through the expected life of the Project or through 2050 (e.g., ongoing planning efforts and protective measures in place, planned, or proposed). If the Discharger has not yet established a plan, explain the process and timeline for doing so. If the Discharger believes it will not be susceptible to flooding related to groundwater rise within 50 years, explain the basis for the conclusion.

The plan shall propose an adaptive management strategy that will be updated every five years and compared against actual sea-level rise and updated projections.

Climate change is shifting precipitation and temperature patterns, exacerbating extreme weather events, and causing sea level and groundwater rise. These conditions have significant implications for the Project.

We consider the Ocean Protection Council's *Sea-Level Rise Guidance*⁶ to be the most current authoritative source supporting planning for sea level rise in California. In May 2020, the California Coastal Commission adopted *Making California's Coast Resilient to Sea Level Rise: Principles for Aligned State Action*.⁷ The California Environmental Protection Agency, including the State Water Resources Control Board, has endorsed these principles, which recommend using a minimum sea level rise target of 3.5 feet by 2050 for planning purposes. This target applies a safety factor to the California Ocean Protection Council's sea level rise estimates, which do not account for extreme storm surges, tides, or other weather events on top of sea level rise.

The Flood Protection Plan may evaluate and select strategies consistent with the most recent version of the San Francisco Bay Shoreline Adaptation Atlas (currently 2019), prepared by the San Francisco Estuary Institute (Adaptation Atlas). The Adaptation Atlas serves as an important science-based tool for developing adaptation strategies for the Bay shoreline as climate change impacts the shoreline. The Adaptation Atlas uses a framework of Operational Landscape Units (OLUs) where the key purpose of the OLU framework is to identify where it may be possible to use nature-based approaches, such as beaches, marshes, and subtidal reefs, to create a resilient shoreline with multiple benefits. Nature-based approaches, and hybrid measures that integrate nature with engineered structural approaches, may perform better than traditional engineered infrastructure alone.

VIII. ENFORCEMENT

Please review this NOA carefully to ensure that it completely and accurately reflects the discharge of biosolids to land for use as a soil amendment in agricultural activities. Failure to abide by the conditions of the General Order and this NOA could result in enforcement actions, as authorized by provisions of the Water Code.

Prior to implementing any discharge changes, a new NOI must be submitted for continued coverage under the General Order. Alternatively, a Report of Waste Discharge may be submitted for coverage under individual Waste Discharge Requirements.

IX. ANNUAL FEES

The Discharger is responsible for payment of an annual fee, pursuant to Water Code section 13263 and the Annual Fee Schedule as stated in the California Code of Regulations, title 23, division 3, chapter 9, section 2200. The fee amount is determined by the type of order and the threat to water quality and complexity rating. Biosolids

⁶ https://opc.ca.gov/webmaster/ftp/pdf/agenda_items/20180314/Item3_Exhibit-A_OPC_SLR_Guidance-rd3.pdf

⁷ <https://documents.coastal.ca.gov/reports/2020/5/w6g/w6g-5-2020-exhibits.pdf>

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application projects greater than or equal to 40 acres are deemed as a category "2" threat to water quality rating and a category "B" complexity rating. The corresponding annual fee is currently **\$23,783** (2021-22 Fee Schedule). The Water Quality Fee Schedule is subject to change annually and is accessible online at http://www.waterboards.ca.gov/resources/fees/water_quality/#wdr.

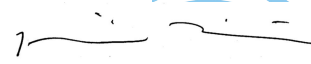
The fee is due and payable on an annual basis until coverage under the General Order is formally terminated. Coverage under the General Order may be terminated by submittal of the Final Monitoring and Reporting Program technical report and a Notice of Termination (NOT). For sites using Class B biosolids, termination shall not take place until 38 months after the last Class B biosolids application. The Discharger will be responsible for paying all annual fees until approval of the NOT is granted by the Water Board's Executive Officer. If an individual Waste Discharge Requirements (WDR) Order is issued for this Project, the applicability of this General Order to the Discharger is automatically terminated on the effective date of the individual WDR Order.

X. AUTHORIZATION FOR COVERAGE UNDER ORDER 2004-0012-DWQ

I hereby approve the Project as described in the Notice of Intent and supplemental information to henceforth be regulated under the General Order. Failure to abide by the conditions of the General Order, this Notice of Applicability, and the Monitoring and Reporting Program, could result in enforcement actions, as authorized by provisions of the California Water Code.

If you have any questions or would like to discuss further, contact Margaret Monahan of my staff via email to Margaret.Monahan@waterboards.ca.gov or at (510) 622-2377.

Sincerely,


Digitally signed by
Keith H. Lichten,
Division Chief
Date: 2022.07.01
09:58:26 -07'00'


for Thomas Mumley
Interim Executive Officer

Attachments: 1. Monitoring and Reporting Program
 2. General Order 2004-0012-DWQ
 3. Project Location and Biosolids Application Area Maps

Copy to (by email): Las Gallinas Valley Sanitary District
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ATTACHMENT 1: Monitoring and Reporting Program

**SAN FRANCISCO BAY
REGIONAL WATER QUALITY CONTROL BOARD**

**MONITORING AND REPORTING PROGRAM
for
LAS GALLINAS VALLEY SANITARY DISTRICT BIOSOLIDS
MARIN COUNTY**

I. MONITORING AND REPORTING PROGRAM OVERVIEW

This Monitoring and Reporting Program (Monitoring Program) describes requirements for monitoring biosolids and biosolids disposal and land application areas for Las Gallinas Valley Sanitary District (Discharger).

The Discharger is enrolled under General Waste Discharge Requirements for the Discharge of Biosolids to Land for Use as a Soil Amendment in Agricultural, Silvicultural, Horticultural, and Land Reclamation Activities, Order 2004-0012-DWQ (General Order) via a Notice of Applicability (NOA) letter. The enrollment conditionally authorizes the land application of biosolids, generated from the Discharger's wastewater treatment plant at 300 Smith Ranch Road, San Rafael, in Marin County (Facility), to agricultural areas on District property. The Discharger is responsible for compliance with the monitoring and reporting requirements specified in this Monitoring Program.

Monitoring and reporting responsibilities of waste dischargers are specified in California Water Code (Water Code) sections 13267(b), 13268, and 13383, which authorize the San Francisco Bay Regional Water Quality Control Board (Water Board) to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. This Monitoring Program is issued pursuant to Water Code sections 13383 and 13267.

The monitoring reports are necessary to document the Discharger's compliance with biosolids management requirements including Title 40 of the Code of Federal Regulations (CFR), Part 503, the General Order, and the NOA, and the burden imposed in their preparation bears a reasonable relationship to the need for the reports and the benefits to be obtained from them. More specifically, the monitoring data are needed to evaluate biosolids disposal and land application practices to assure protection of public health and the environment. Mismanagement of biosolids could expose people to pathogens or other harmful constituents and could adversely affect beneficial uses of underlying groundwater, adjacent tidal wetlands, and adjacent surface waters including the San Pablo Bay. The biosolids disposal and land application locations are adjacent to the San Pablo Bay to the east and are separated from tidal action by constructed levees along the east and south boundaries.

Pursuant to Water Code sections 13383 and 13267, the Discharger shall implement this Monitoring Program and submit the monitoring reports described herein.

II. MONITORING AND REPORTING PROGRAM CHANGES

1. The Water Board Executive Officer may authorize changes to the monitoring and reporting practices specified in this Monitoring Program, in consideration of acceptable alternate means of monitoring, and the total inventory of monitoring data. Factors to be considered include the following: data quality, adequate characterization, compliance with requirements, and acceptable means for providing equivalent and adequate monitoring.
2. Requests for changes to monitoring frequencies, parameters, or reporting practices must be submitted to the Regional Water Board in writing, with an evaluation of accumulated data, complete description of proposed changes, and alternate means of monitoring.
3. The Discharger shall not implement changes to authorized monitoring and reporting practices until applicable changes are authorized in writing by the Executive Officer.

III. QUALITY ASSURANCE AND CONTROL

1. All samples shall be representative of the volume and nature of the discharge or matrix of material sampled. The Discharger shall use clean sample containers and sample handling, storage, and preservation methods in accordance with approved U.S. EPA analytical methods or as recommended by the selected analytical laboratory. All analytical samples shall be labeled and records maintained to show the name of the sampler, date, time, sample location, sample type, collection method, bottle type, and any preservative used for each sample. All samples collected for laboratory analyses shall be preserved as required and submitted to the laboratory within the required holding time appropriate for the analytical method used and the constituents analyzed.
2. All samples submitted to a laboratory for analysis shall be identified in a properly completed and signed chain of custody form containing the sampler, date, time, sample location, sample type, collection method, bottle type, and any preservative used for each sample. The chain of custody form shall also contain custody information, including the date, time, transport method, and to whom samples were relinquished.
3. Consistent with Water Code section 13176, data produced and reports submitted for compliance with the General Order must be generated by a laboratory with accreditation from the State Water Board, Division of Drinking Water, Environmental Laboratory Accreditation Program (ELAP), where accreditation is specific to the analyses required, or the laboratory must hold a valid certificate of accreditation for equivalent analytical test methods validated for the intended uses and approved by the State Water Board or regional water board. The laboratory must include quality assurance/quality control data in all data reports and submit electronic data as required by the State Water Board and regional water boards. Where technically feasible, laboratory reporting limits shall be lower than concentrations that implement applicable water quality objectives or limits for the constituents to be analyzed.

4. Monitoring must be conducted according to test procedures approved under 40 CFR part 136 for the analyses of pollutants unless another method is required under 40 CFR chapter 1, subchapter N. Monitoring must be conducted according to sufficiently sensitive test methods approved under 40 CFR part 136 for the analysis of pollutants or pollutant parameters or as required under 40 CFR chapter 1, subchapter N. For the purposes of this paragraph, a method is sufficiently sensitive when:
 - a. The method minimum level (ML) is at or below the level of the most stringent effluent limitation established in the permit for the measured pollutant or pollutant parameter, and either the method ML is at or below the level of the most stringent applicable water quality criterion for the measured pollutant or pollutant parameter or the method ML is above the applicable water quality criterion but the amount of the pollutant or pollutant parameter in the facility's discharge is high enough that the method detects and quantifies the level of the pollutant or pollutant parameter in the discharge; or
 - b. The method has the lowest ML of the analytical methods approved under 40 CFR part 136 or required under 40 CFR chapter 1, subchapter N, for the measured pollutant or pollutant parameter.

In the case of pollutants or pollutant parameters for which there are no approved methods under 40 CFR part 136 or otherwise required under 40 CFR chapter 1, subchapter N, monitoring must be conducted according to an appropriate test procedure for such pollutants or pollutant parameters. (40 CFR parts 122.21(e)(3), 122.41(j)(4), 122.44(i)(1)(iv).)

5. Data generated using field tests are exempt pursuant to California Water Code Section 13176. Field instruments may be used to test field parameters (such as for pH, electrical conductivity, and dissolved oxygen) provided that the operator is trained in the proper use of the instrument and each instrument is serviced and/or calibrated at the recommended frequency by the manufacturer and in accordance with manufacturer instructions. Field calibration reports shall be maintained for at least 3 years.
6. All sample and analysis field logs, laboratory reports, and quality assurance/quality control data shall be reported with the sample results to which it applies. The reports shall include applicable information such as the method, equipment, analytical detection, quantitation limits, recovery rates, an explanation for any recovery rate that is outside method specifications, results of method blanks, results of matrix spikes and surrogate samples, and the frequency of quality control analysis. Sample results shall be reported unadjusted for blank results or spike recovery. In cases where contaminants are detected in the quality assurance/quality control samples (e.g., laboratory blanks), the accompanying sample results shall be appropriately flagged.

IV. MONITORING STATIONS

1. **Monitoring Station Definitions.** Stations to be used for sampling and observations in accordance with this Monitoring Program shall be described in each monitoring report submitted in response to this Monitoring Program. The station description shall include the matrix being monitored at that station and the location.
2. **Monitoring Station Changes.** Changes to approved monitoring stations may be authorized by the Executive Officer in order to accommodate changes in operations or to provide improved monitoring. Requests for changes to the monitoring stations must be submitted to the Water Board in writing, with a detailed explanation of the purpose of the proposed station changes. Proposed changes to monitoring stations must be approved in writing from the Executive Officer prior to implementation.
3. **Site Plan Showing All Monitoring Stations.** The Discharger shall develop a legible plan view drawing that clearly depicts the locations of all biosolids management areas, all monitoring stations, and relevant land use features such as buildings, access roads, property boundaries, and surface water drainage systems. A copy of this drawing shall be included with all reports submitted in response to this Monitoring Program.

V. MONITORING REQUIREMENTS

A. BIOSOLIDS MONITORING

The Discharger shall sample and analyze biosolids as follows. Results for chemical constituents shall be reported in milligrams per kilogram (mg/kg) on a dry-weight basis. Whenever possible and appropriate, composite sampling should be conducted (e.g., for metals).

Table 1. Biosolids Monitoring

Constituent	Units ¹	Sample Frequency
EPA Priority Pollutant List	mg/kg, µg/kg	Once every five years
Polycyclic Aromatic Hydrocarbons (PAHs)	µg/kg	Once every five years
Volatile Organic Compounds (VOCs)	µg/kg	Once every five years
Metals ²	mg/kg	Once per year
Cyanide	mg/kg	Once per year
PCB arochlors, aldrin, dielfrin ³	µg/kg	Once per year
Semi-volatile organics (SVOCs) ⁴	µg/kg	Once per year
Total nitrogen	mg/kg	Once per year
Ammonia nitrogen (as N)	mg/kg	Once per year

Constituent	Units ¹	Sample Frequency
Nitrate nitrogen (as N)	mg/kg	Once per year
Total phosphorous (as P)	mg/kg	Once per year
Total potassium	mg/kg	Once per year
pH	s.u.	Once per year
Salinity	mg/kg	Once per year
Total solids content	%	Once per year
Percent moisture	%	Once per year

¹ mg/kg = milligram per kilogram; µg/kg = microgram per kilogram; MPN/100 g = most probable number per 100 grams; s.u. = standard units; % = percent

² Metals include arsenic, cadmium, chromium, copper, lead, mercury, molybdenum, nickel, selenium, silver, and zinc.

³ PCB arochlors, aldrin, dielfrin: The General Order specifies SW 846 Method 8080.

⁴ SVOCs: The General Order specifies EPA Method 8270.

Monitoring for additional constituents may be warranted based on the Biosolids and Site Characterization Report.

B. ROUTINE FIELD MONITORING

The Discharger shall establish and implement an inspection and application oversight program to monitor and control biosolids application rates and ensure compliance with the NOA and General Order. Each discrete application field shall be managed and monitored as follows:

1. Pre-application Oversight:
 - a. Define crop to be planted.
 - b. Calculated allowable loading rate based on soil nitrogen residual data from the previous fall and most recent plant available nitrogen (PAN) and moisture content data.
 - c. Document communication of allowable loading rates to spreader operator.
2. Pre-application Inspection:
 - a. Verify that setbacks are clearly delineated.
 - b. Verify that runoff controls are in place and functional.
 - c. Verify that culverts are blocked (where applicable).
3. Application Oversight:
 - a. Verify compliance with setbacks and allowable loading rate.
 - b. Verify compliance with soil incorporation requirements.

4. Post-application Oversight:

- a. Confirm with irrigation manager requirements to control runoff for the specified period after application.
- b. Calculate actual biosolids and PAN loading rates.
- c. Note anticipated dates of planting, irrigation, and harvest.

C. SOIL MONITORING

The Discharger shall establish an annual soil sampling program as follows: two background sampling locations outside of the land application areas (e.g., within application setback areas) and, at least six sampling locations within each discrete land application area identified in the NOA. Sampling locations shall be distributed to be representative of each subarea and predominant soil type.

Soil samples shall be collected from each sampling location at the following depths intervals: 0 to 3 inches, 4 to 12 inches, and 13-24 inches below ground surface. The top organic layer of soil must be removed prior to taking sample. Each sample shall be thoroughly mixed to create a composite sample representative of the depth interval and shall be analyzed as specified in the following table. Annual samples shall be collected in the fall (fourth quarter) and must occur at the same time each year.

Table 2. Soil Monitoring

Constituent/Parameter	Units ¹	Sample Frequency
Soil Classification (USCS and USDA)	--	Annually
Total Solids	%	Annually
Total Alkalinity ²	mg/kg as CaCO ₃	Annually
Cation Exchange Capacity ²	meq/100 grams	Annually
Electrical Conductivity	mg/kg, mg/L	Annually
Chloride ³	mg/L	Annually
pH	s.u.	Annually
Metals ^{2, 3, 4}	mg/kg, mg/L	Annually

¹ % = percent; mg/kg = milligrams per kilogram; CaCO₃ = calcium carbonate; meq/100 grams = milliequivalents per 100 grams soil; mg/L = milligrams per liter; s.u. = standard units

² For specified constituents, results shall be reported on a dry-weight basis; show calculations.

³ For specified constituents, analysis shall be performed on the extract obtained from the Waste Extraction Test using distilled water as the extractant.

⁴ Metals include arsenic, cadmium, chromium, copper, lead, mercury, molybdenum, nickel, selenium, silver, and zinc.

D. GROUNDWATER MONITORING

The groundwater monitoring program applies to the groundwater monitoring wells tabulated below and any wells subsequently installed under the Water Board's direction.

Table 3. Groundwater Monitoring Well Network

Parcel	Monitoring Well
155-011-33	MW-01
155-011-33	MW-02
155-011-33	MW-03
155-011-33	MW-04
155-011-14	MW-05
155-011-14	MW-06
155-011-14	MW-07
155-011-14	MW-08
155-011-13	MW-09
155-011-13	MW-10
Dedicated Land Disposal Area	G-1
Dedicated Land Disposal Area	G-3
Dedicated Land Disposal Area	G-4
Dedicated Land Disposal Area	G-5

On July 27, 2021, the Discharger submitted a *Monitoring Well Installation Work Plan* (Work Plan) to the Water Board for review and approval. The Water Board issued comments on the Work Plan on December 15, 2021. A revised Work Plan must be approved prior to Work Plan implementation. Upon completion and development of groundwater monitoring wells installed in accordance with an approved Work Plan, the Discharger shall implement the following groundwater monitoring program for all monitoring wells. Once installed, all new wells shall be added to this Monitoring Program and all wells shall be sampled and analyzed according to the schedule below.

Prior to purging, groundwater elevations shall be measured, and the wells shall be purged of at least three well volumes until temperature, pH, and electrical conductivity have stabilized prior to sampling. Depth to groundwater shall be measured to the nearest 0.01 feet. Water table elevations shall be calculated and used to determine groundwater gradient and direction of flow. Samples shall be collected using approved EPA methods. Groundwater monitoring shall include, at a minimum, constituents specified in the table below. Groundwater elevation shall be determined on depth to

water measurements using a surveyed measuring point elevation on the well and a surveyed reference elevation.

Table 4. Groundwater Monitoring

Constituent/Parameter	Units¹	Type of Sample	Sample Frequency²
Depth to groundwater	0.01 feet	Measurement	Quarterly for first year; semi-annually thereafter
Groundwater elevation	0.01 feet	Calculated	Quarterly for first year; semi-annually thereafter
Gradient magnitude	feet/feet	Calculated	Quarterly for first year; semi-annually thereafter
Gradient direction	Degrees	Calculated	Quarterly for first year; semi-annually thereafter
pH	s.u.	Grab	Quarterly for first year; semi-annually thereafter
Electrical conductivity	µmhos/cm	Grab	Quarterly for first year; semi-annually thereafter
Chloride	mg/L	Grab	Quarterly for first year; semi-annually thereafter
Nitrate as Nitrogen	mg/L-N	Grab	Quarterly for first year; semi-annually thereafter
Nitrite as Nitrogen	mg/L-N	Grab	Quarterly for first year; semi-annually thereafter
Sulfate	mg/L	Grab	Quarterly for first year; semi-annually thereafter
Metals ³	mg/L	Grab	Quarterly for first year; semi-annually thereafter
Alkalinity	mg/L as CaCO ₃	Grab	Quarterly for first year; semi-annually thereafter
Total Dissolved Solids (TDS)	mg/L	Grab	Quarterly for first year; semi-annually thereafter
Total Kjeldahl Nitrogen (TKN)	mg/L	Grab	Quarterly for first year; semi-annually thereafter
Total Phosphorous	mg/L	Grab	Quarterly for first year; semi-annually thereafter
Sulfide	mg/L	Grab	Quarterly for first year; semi-annually thereafter
Cyanide	mg/L	Grab	Quarterly for first year; semi-annually thereafter
Ammonia	mg/L	Grab	Quarterly for first year; semi-annually thereafter
Total Organic Carbon (TOC)	mg/L	Grab	Quarterly for first year; semi-annually thereafter

Constituent/Parameter	Units ¹	Type of Sample	Sample Frequency ²
Total Nitrogen	mg/L	Calculated	Quarterly for first year; semi-annually thereafter
Fecal coliform ⁴	MPN/100 mL	Grab	Quarterly for first year; semi-annually thereafter

¹ s.u. = standard units; $\mu\text{mhos/cm}$ = micromhos per centimeter; mg/L = milligrams per liter; CaCO_3 = calcium carbonate

² Semi-annual samples should be collected at the end of the dry and wet seasons.

³ Metals include arsenic, cadmium, chromium, copper, lead, mercury, molybdenum, nickel, selenium, silver, and zinc.

⁴ Fecal coliform reported as most probable number of colony-forming units per 100 milliliters

VI. REPORTING REQUIREMENTS

All monitoring reports should be converted to a searchable Portable Document Format (PDF) and submittal electronically. All electronic files shall be submitted via the Water Board's centralized WDR monitoring report email address:

WDR.Monitoring@waterboards.ca.gov.

A transmittal letter shall accompany each monitoring report. The letter shall include a discussion of all violations during the reporting period and actions taken or planned for correction of each violation. If the Discharger has previously submitted a report describing corrective actions taken and/or a time schedule for implementing the corrective actions, reference to the previous correspondence will be satisfactory. The transmittal letter shall contain a statement by the Discharger or the Discharger's authorized agent certifying under penalty of perjury that the report is true, accurate, and complete to the best of the signer's knowledge.

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., biosolids, soil, ground water monitoring, etc.) and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in this Monitoring Program shall be reported in the next scheduled monitoring report.

Laboratory analysis reports need to be included in the monitoring reports; all laboratory reports must be retained for a minimum of three years. For a discharger conducting any of its own analyses, reports must also be signed and certified by the chief of the laboratory.

Monitoring information shall include the method detection limit (MDL) and the reporting limit (RL) or practical quantitation limit (PQL). If the regulatory limit for a given constituent is less than the RL (or PQL), then any analytical results of that constituent that are below the RL (or PQL) but above the MDL shall be reported and flagged as estimated.

All monitoring reports that involve planning, investigation, evaluation or design, or other work requiring interpretation and proper application of engineering or geologic sciences, shall be prepared by or under the direction of persons registered to practice in California pursuant to California Business and Professions Code section 6735, 7835, and 7835.1.

A. PRE-APPLICATION REPORT

A Pre-Application Report shall be submitted for each field prior to the application of biosolids in accordance with the General Order and this Monitoring Program. A Pre-Application Report shall be submitted at least 30 days prior to the date of the proposed application. The Pre-Application Report shall be signed by the Discharger (in this case the Discharger is both the owner/operator of the biosolids application operation and the property owner).

B. ANNUAL REPORT

An Annual Report shall be prepared and submitted by February 1 each year. The Annual Report shall include the following:

1. A summary of all analytical data and verification of compliance with the biosolids monitoring requirements.
2. For each discrete application field, a chronological log of dates of biosolids application, irrigation, precipitation, and runoff control operations. Specifically, information demonstrating compliance with the routine field monitoring requirements.
3. For each discrete application field:
 - a. Total cumulative metals loading rates as of the end of the previous calendar year;
 - b. Calculation of the total metals and nitrogen loading rates for the year;
 - c. The cumulative metals loading rates since biosolids land application began; and
 - d. The cumulative metals loading rates to date as a percentage of the cumulative metals loading limits.
4. A report of soil monitoring, including:
 - a. Sampling and analysis activities, including a scaled map of sampling locations;
 - b. Tabulation of all soil analytical results;
 - c. Historical time vs. concentration plots for each constituent at each sampling interval;
 - d. A discussion of any observed spatial or temporal variation; and
 - e. Whether pH adjustment is needed and, if so, how and when the adjustment will be made.

5. A groundwater monitoring summary report including:
 - a. Tabular and graphical summaries of all data collected during the year;
 - b. An evaluation of the groundwater quality beneath the site;
 - c. A discussion of compliance and the corrective actions taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements;
 - d. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program; and
 - e. The results for all groundwater analyses conducted in the year.
6. A copy of calibration log page(s) verifying calibration of all hand-held monitoring instruments performed during the year.
7. For the duration of the research project *Unregulated Organic Chemicals in Biosolids: Prioritization, Fate and Risk Evaluation for Land Applications*, EPA Grant Number R840245, for which the Discharger is a participating site, the annual report shall include a status update on research project work associated with the Discharger site.

MONITORING PROGRAM CERTIFICATION

The Discharger shall implement this Monitoring and Reporting Program as of the effective date given below.

I, Thomas Mumley, Interim Executive Officer, hereby certify that this Monitoring and Reporting Program:

1. Has been developed in accordance with the procedure set forth in the Board's Resolution No. 73-16 in order to obtain data and document compliance with waste discharge requirements for the subject wastewater systems.
2. May be reviewed at any time subsequent to the effective date upon written notice from the Executive Officer or request from the Discharger, and revisions will be ordered by the Executive Officer.
3. Is effective on the following date: July 1, 2022.



for Thomas Mumley
Interim Executive Officer

ATTACHMENT 2: General Order 2004-0012-DWQ

**STATE WATER RESOURCES CONTROL BOARD
WATER QUALITY ORDER NO. 2004 - 0012 - DWQ**

**GENERAL WASTE DISCHARGE REQUIREMENTS FOR THE
DISCHARGE OF BIOSOLIDS TO LAND FOR USE AS A SOIL
AMENDMENT IN AGRICULTURAL, SILVICULTURAL,
HORTICULTURAL, AND LAND RECLAMATION ACTIVITIES
(GENERAL ORDER)**

The State Water Resources Control Board (hereinafter referred to as the SWRCB) finds that:

1. Applications for the use of treated municipal sewage sludge meeting the requirements specified in Part 503 in Title 40 of the Code of Federal Regulations (CFR) (hereinafter referred to as biosolids) as a soil amendment have been received and waste discharge requirements (WDRs) have been issued by several of the nine Regional Water Quality Control Boards (RWQCBs). Section 13274 of the California Water Code (CWC) requires the SWRCB or RWQCBs to prescribe General WDRs for the discharge of biosolids used as a soil amendment. This General Order is intended to satisfy the requirements of CWC section 13274 and is intended for discharges of biosolids for use as a soil amendment. This General Order assists in streamlining the regulatory process for such discharges but may not be appropriate for all sites using biosolids due to particular site-specific conditions or locations. Such sites are not precluded from being issued individual WDRs. For the purposes of this General Order, biosolids do not include septage. Biosolids material applicable for coverage under this General Order is as described below:
 - a. All Class A biosolids not meeting the requirements contained in Table 3 of 40 CFR Part 503.13 and Class B biosolids that are land applied for agricultural, silvicultural, horticultural, and land reclamation activities;
 - b. All Exceptional Quality (EQ) biosolids-derived mixtures consisting of more than or equal to 50 percent biosolids (dry weight) applied at more than 10 dry-tons per acre per year for use as a soil amendment to continuous fields/plots greater than 20 acres for agricultural, silvicultural, horticultural, and land reclamation activities and where the said fields/plots are owned or operated by the same person, company, or partnership;
 - c. All EQ biosolids-derived mixtures consisting of 50 percent biosolids or less (dry weight) applied at more than 20 dry-tons per acre per year for use as a soil amendment to continuous fields/plots greater than 20 acres for agricultural, silvicultural, horticultural, and land reclamation activities and where the said fields/plots are owned or operated by the same person, company, or partnership.
2. EQ biosolids may not necessitate regulation in the future. However, it is believed that large scale uses currently require oversight regardless of the actual threat to water quality while done at agronomic rates and using best management practices. Accordingly, this General Order can be applied to such sites to ensure that biosolids are being properly used and are not used in an activity of unregulated

dumping. This regulatory tool may be used to regulate material that is land applied at a high loading rate in order to discourage poor biosolids management and to reduce risk to the public and the environment.

3. Within this General Order, the following terms are described as follows:
 - a. Agriculture: The practice, science, or art of using the soil for the production of crops and/or raising livestock for human use.
 - b. Agricultural Mineral: Any material containing nitrogen, available phosphoric acid, or soluble potash, singly or in combination, in amounts less than 5 percent or any substance containing essential secondary nutrients or micronutrients that is distributed for use in agriculture, silviculture, horticulture, and land reclamation activities for the purpose of promoting plant growth.
 - c. Agronomic Rate: The nitrogen requirements of a plant needed for optimal growth and production, as cited in professional publications for California or recommended by the County Agricultural Commissioner, a Certified Agronomist or Certified Soil Scientist.
 - d. Applier: Person, group of persons, or company that applies biosolids for use as a soil amendment.
 - e. Arid: Arid lands are those areas where the long term annual average rainfall is below 250 millimeters (less than 10 inches).
 - f. Biosolids: Sewage sludge that has been treated and tested and shown to be capable of being beneficially and legally used as a soil amendment for agriculture, silviculture, horticulture, and land reclamation activities as specified under 40 CFR Part 503.
 - g. Buffer Zones: An area of land that provides a separation distance between the land application site and an area of concern.
 - h. Class A Biosolids: Biosolids meeting the vector attraction, and meeting pollution concentration limits specified in 40 CFR Part 503 and pathogen reduction standards specified in 40 CFR Part 503.32(a).
 - i. Class B Biosolids: Biosolids meeting the vector attraction and meeting pollution concentration limits specified in 40 CFR Part 503 and pathogen reduction standards specified in 40 CFR Part 503.32(b).
 - j. Depth to Ground Water: The distance from the land surface elevation to the seasonal high water table.
 - k. Domestic Water Supply Well: A well that provides water used for human consumption.
 - l. EQ Biosolids: Biosolids which meet metals standards, Class A pathogen reduction standards, and vector attraction reduction standards contained in

40 CFR Part 503.13 (Table 3), 40 CFR Part 503.32, and 40 CFR Part 503.33, respectively.

- m. Fallow: Fallow lands are areas that have not been cultivated during the growing season but do not include areas that have been tilled, disked, or otherwise distributed to control weeds or conserve soil moisture during such season.
- n. Fertilizing Material: Biosolids with 5 percent or more of nitrogen, available phosphoric acid, or soluble potash, singly or in combination.
- o. Generator: Municipal Wastewater Treatment Facility or Sewage Sludge Treatment Facility.
- p. Grower: Person or entity primarily responsible for planting, maintaining, and harvesting or allowing the use of crops and/or range land for domestic animal or human use.
- q. Gully erosion: Erosion cut by a concentrated but intermittent flow of water usually during and immediately following heavy rains or after ice/snow melt. A gully generally is an obstacle to wheeled vehicles and too deep (e.g., > 0.5 meter) to be obliterated by ordinary tillage.
- r. High Potential for Public Exposure Areas: Land located within one-half mile of educational facilities, facilities designated for recreational activities other than hunting, fishing, or wildlife conservation, places of public assembly, hospitals, or similar sensitive receptors.
- s. Horticulture: The practice, science, or art of cultivating the soil to produce fruit, vegetables, or ornamental plants for human use.
- t. Key Operating Personnel: Those individuals responsible for the oversight of daily operations, management decisions, and planning of biosolids land application projects.
- u. Low Potential for Public Exposure Areas: Land not meeting the definition of High Potential for Public Exposure Areas.
- v. Label: The display of all written, printed, or graphic matter on the immediate container of, or a statement including the guaranteed analysis, accompanying fertilizing material as required by the California Department of Food and Agriculture.
- w. Land Reclamation: The practice of revitalizing or restoring lands that are damaged from past or present human land use practices.
- x. Long-Term Storage Facility: Site which holds biosolids for more than seven days consecutively.
- y. Micronutrients: Refers to boron, chloride, cobalt, copper, iron, manganese, molybdenum, sodium, or zinc.

- z. Municipal Wastewater Treatment Facilities (treatment facilities): Facilities designed to collect and treat wastewater generated from primarily domestic sources for environmentally safe reuse or disposal.
- aa. Notice of Applicability: Written notice that a biosolids land application site is required to comply with the provisions of this General Order and that applications according to the General Order may commence.
- ab. Notice of Intent (NOI): Application for coverage under this General Order, as attached. The NOI is also a notification form for the public and interested parties for this General Order.
- ac. Notice of Termination (NOT): Request form to discontinue coverage of this General Order.
- ad. Nuisance: Nuisance means anything which meets all of the following requirements:
 - (1) Is injurious to health, or is indecent and offensive to the sense, or is an obstruction to the free use of property so as to interfere with the comfortable enjoyment of life and property.
 - (2) Affects at the same time an entire community or neighborhood or any considerable number of persons, although the extent of the annoyance or damage inflicted upon individuals may be unequal.
 - (3) Occurs during, or as a result of, the treatment or disposal of wastes.
- ae. Pathogens: Disease causing agents including helminths, bacteria, viruses, and protozoa.
- af. Pathogen Reduction: Process used to destroy pathogenic material contained in sewage sludge.
- ag. Pollution: Means an alteration of the quality of the waters of the State by waste to a degree which unreasonably affects either of the following:
 - (1) The waters for beneficial uses.
 - (2) Facilities which serve these beneficial uses.
- ah. Secondary Nutrients: The elements of calcium, magnesium, and sulfur.
- ai. Septage: Waste material removed from a septic tank, cesspool, portable toilet, Type III marine sanitation device, or similar wastewater handling device that has not passed through a municipal wastewater treatment facility.
- aj. Sewage Sludge: The solid, semisolid, or liquid residue generated during the treatment of domestic sewage in a municipal wastewater treatment facility. Sewage sludge includes solids removed or used during primary, secondary, or advanced wastewater treatment processes. Sewage sludge does not include grit

or screening material generated during preliminary treatment of domestic sewage at a municipal wastewater treatment facility.

- ak. Short-Term Storage: Biosolids storage sites used as a temporary holding facility for less than or equal to seven days.
 - al. Silviculture: The practice, science, or art of managing, developing, and harvesting forests and trees for human use.
 - am. Soil Amendment: Applications of a fertilizing material or agricultural mineral for the purpose of promoting utilization by plants and other living organisms with the goal of a net gain in soil productivity.
 - an. Staging Area: Area used to hold biosolids for less than 48 hours prior to use for the specified activity listed in the NOI.
 - ao. Tailwater: Excess water from crop irrigation resulting in a discharge off site to a surface water body.
 - ap. Vector Attraction: Characteristic of biosolids that attracts potential pathogen transmitters such as flies, rodents, and other animals or organisms.
 - aq. Water-saturated soil: Water content of the soil such that any further addition of water will result in runoff, standing water, or percolation of water through the displacement of existing soil water.
4. Treatment facilities serve urban and suburban population areas by collecting and treating municipal wastewater and reusing or disposing of wastewater effluent. While serving the public in this manner, significant amounts of sewage sludge are generated. This material is typically further treated (stabilized) and dewatered resulting in biosolids as a product of the wastewater treatment process. Biosolids can be managed using a variety of options including: (a) disposal in a sanitary landfill, (b) incineration, (c) placement into a landfill dedicated for this purpose, (d) use as daily landfill cover, and (e) use in land application operations, including reclamation, horticulture, agriculture, and silviculture. As population increases and technological improvements in wastewater treatment processes occur, the amount of biosolids generated in California is likely to increase significantly.
5. Particularly in urban areas, industrial sources discharge into wastewater collection systems. Many of these discharges are regulated by pretreatment programs implemented pursuant to 40 CFR Part 403. These programs restrict industries from discharging toxic pollutants in concentrations creating concerns for the treatment facilities.
6. As a result of domestic and industrial uses, pollutants enter the collection system of treatment facilities. The majority of the pollutant load treated at the treatment facilities is organic matter. This material is removed through flotation and/or settling or is converted to biological solids and then removed through settling prior to discharge. The settled material is then further treated to stabilize organic matter which constitutes the majority of the domestic sewage sludge. Metals from domestic and industrial sources are also present in the waste stream at the treatment facility. These pollutants are removed from the waste stream and concentrated in

the sewage sludge. Organic chemicals can also be present from domestic and industrial uses of water. The fate of these pollutants is variable. Some are removed and destroyed through physical and biological processes at the treatment facility. Others may concentrate in the sewage sludge. Some pass through the treatment facilities unchanged and are subsequently discharged from the treatment process. A portion of the organic chemicals concentrated in the sewage sludge is degraded during sludge stabilization processes. Some organic chemicals can remain in the sewage sludge unchanged. For these reasons, testing of sewage sludge is necessary prior to it being classified as biosolids.

7. Biosolids are a source of organic matter, nitrogen, phosphorus, and micronutrients. These materials are beneficial to agriculture, silviculture, horticulture, and land reclamation activities and they improve agricultural productivity. More specifically, the benefits derived from biosolids used as a soil amendment are as follows:
 - a. Nitrogen is a basic nutrient for plant growth. In biosolids, it is present in the forms of ammonia, nitrates, and organic nitrogen at concentrations from two to 10 percent by weight on a dry weight basis. The ammonia and nitrate forms of nitrogen are available for plant usage. Organic nitrogen is released slowly (mineralized) over many months, providing a continuous supply of nitrogen for crops and minimizing the potential for movement of nitrogen to the ground water. Ammonium and nitrate (and some nitrite) are the available forms of nitrogen that are taken up by the plants and some form salt reserves and mineralized organic nitrogen in the soil. Total nitrogen available to the plant at any given time is less than the total of these mineral forms due to the dynamic cycling of nitrogen in the soil.
 - b. Phosphorus is a basic nutrient for plant growth and is present in all biosolids in varying concentrations.
 - c. Micronutrients, including a variety of salts and metals, are necessary for plant growth and are present in biosolids in varying amounts.
 - d. The addition of biosolids to soils can also be beneficial by enhancing soil structure, increasing water retention capability, promoting soil aggregation, and reducing the bulk density. Organic matter assists in maintaining soil pores which allow water and air to pass through the soil medium. Such pores can be lost at sites under continuous cultivation and they are critical in maintaining an aerobic environment within the plant root zone.
 - e. Organic matter helps soils retain water. Additional water retention can reduce the need for frequent water applications and can facilitate water conservation in the soil column.
 - f. Liming agents are available when the biosolids have been chemically stabilized with lime. Liming agents increase soil pH and can improve the permeability of the soils. Higher pH soils have a greater propensity to bind most heavy metals, decreasing the chance of the metals migrating to the ground water.

8. Biosolids have the following characteristics which can create water quality and public health problems if improperly treated, managed, and regulated during use as a soil amendment:
 - a. Pathogens can be present. Unless the biosolids are specially treated or disinfected to destroy pathogens, significant concentrations of bacteria, viruses, and parasites can remain. Public health problems can be prevented with appropriate control over public access to the application areas and restrictions on the type and use of crops grown on the application sites. Buffer zones around water supply wells, surface water drainage courses, and public areas are designated to prevent transmission of pathogens to the public.
 - b. Heavy metals will be present. If heavy metals are over-applied to a field, they can cause ground water pollution, toxicity to plants, toxicity/adverse effects to soil microorganisms, or buildup in the plant tissues. A buildup of metals in plant tissues may allow transmission of the metals into the food chain which is the cause of toxicity/adverse effects to animals eating plants or animals containing elevated metals. Future cropping or other land uses could be restricted. Only some of the metals commonly found in biosolids are known to cause water quality or public health problems. Application rates for those metals have been established to avoid the problems.
 - c. Nitrogen can be over-applied, allowing a buildup of nitrogen in soils. Excess nitrogen will eventually be converted to the nitrate form and it can migrate to ground water. Excess nitrate in the ground water can result in the exceedance of drinking water standards and a public health threat. Nitrogen over-application can be prevented by biosolids application at an agronomic rate, that is, by matching the application rate of the nitrogen to the nitrogen usage rate of the crops and to soil permeability and soil retention capability.
 - d. Odor and insect nuisances can be caused if the biosolids have not been adequately treated (stabilized) prior to application or if wet biosolids are allowed to remain on the ground surface for several days. Compliance with State and federal standards for stabilization of the biosolids will minimize the potential for odors and insect nuisances. Proper management at the application site will prevent odor or insect nuisances. Properly stabilized biosolids will generate limited, transient odors in the immediate vicinity of the application operations. Adequate buffer zones around residences and public areas, therefore, should be provided.
 - e. Discharge of organic matter, metals, and pathogens to surface waters can affect water quality. These effects can be prevented by controlling field runoff. The water quality threat of organic matter of biosolids origin affecting surface water is no greater than for a similar quantity of other organic soil amendments.
9. The U. S. Environmental Protection Agency (USEPA) has promulgated 40 CFR Part 503 for the use of biosolids as a soil amendment. These regulations establish ceiling concentrations for metals and pathogen and vector attraction reduction standards; management criteria for the protection of water quality and public health; and annual and cumulative discharge limitations of persistent pollutants, such as heavy metals, to land for the protection of livestock, crop, and

human health and water quality protection. The requirements of 40 CFR Part 503 are based on a risk-based evaluation using 14 different pathways.

10. The National Research Council established a committee to review the methods and procedures used by the USEPA while forming the basis of the 40 CFR Part 503. The National Research Council's members are drawn from the National Academy of Sciences, National Academy of Engineering, and the Institute of Medicine. Committee members included university professors from the schools of law, science, and agriculture; a state health official; a food industry professional; a professional from a sanitation agency; and a professional consultant. After a three-year study (starting in 1993), the committee made some recommendations for improvement of the regulations and data from which they are based but also stated: "Established numerical limits on concentration levels of pollutants added to cropland by sludge are adequate to assure the safety of crops produced for human consumption." As a result of the peer review, monitoring for organic chemicals and using fecal coliform testing as a parameter for determining Class A level pathogen reductions is included in this General Order.
11. This General Order establishes a regulatory system to manage biosolids in a manner that is reasonably protective of public health and the environment to the extent of present scientific knowledge. The beneficial use of biosolids through land application under this General Order is environmentally sound and preferable to non-beneficial disposal.
12. Due to the extensive work done by the USEPA, this General Order is using the 40 CFR Part 503 requirements as baseline requirements for compliance. However, this General Order is applicable to sites where biosolids are applied to land and is not intended to solely regulate the generator (unless the generator is also the landowner or land applier). The 40 CFR Part 503 permit requirements are only intended for and enforceable against the generator. Therefore, this General Order does not constitute compliance with 40 CFR Part 503. Since the SWRCB is not delegated with authority for the Federal Biosolids Program, the USEPA is the only authority to determine compliance with 40 CFR Part 503.
13. Each discharger covered by this General Order shall submit an application fee equal to the annual fee, pursuant to CWC section 13260. The amount of the fee is currently determined by the type of order issued, the threat to water quality, and complexity of the specific discharge, as detailed in Section 2200, Chapter 9, Division 3, Title 23, California Code of Regulations (CCR). Biosolids application projects greater than or equal to 40 acres are deemed as Non-Chapter 15 WDRs with a Category "II" threat to water quality rating and a Category "b" complexity rating. Biosolids projects consisting of less than 40 acres are deemed Category "III" threat to water quality rating and a Category "b" complexity rating.
14. This General Order may be periodically revised to reflect changes in federal or State laws or regulations or policies of the SWRCB or RWQCB.
15. Under CWC section 13263, the SWRCB can prescribe General WDRs for categories of discharges which involve the same or similar waste type or those which are produced by the same or similar operations.

16. This General Order shall primarily apply to both the landowner of sites using biosolids and the biosolids generator, but may also include, as determined by those involved in the operation, the individuals, or companies, transporting and placing the biosolids in the field and the land lessee in conjunction with the landowner and the generator. To obtain coverage under the General Order, a complete NOI and an appropriate fee must be submitted to the RWQCB. Once a completed application is submitted, RWQCB staff will evaluate the project to determine if it is suitable for regulation under this General Order and the corresponding California Environmental Quality Act (CEQA) document. Only after a determination of applicability is made will the discharger be issued a Notice of Applicability by the RWQCB Executive Officer. Only applicants (dischargers) who submit a complete NOI, appropriate fee, and are issued a Notice of Applicability are authorized to land apply biosolids at an agricultural, horticultural, silvicultural, or land reclamation site as a soil amendment onto the land specified in the NOI in compliance with the terms and conditions of this General Order. If it is determined that a local agency already adequately regulates the activity subject to this permit, the RWQCB may choose not to issue this General Order in order to avoid any duplicative regulation.
17. A separate NOI and filing fee must be filed for each biosolids use project to be eligible for coverage under this General Order. A separate NOI and filing fee must be filed for each landowner involved in a reuse project. Attachment A to this General Order contains an NOI form which details the minimum contents of the NOI. A single reuse project will be limited to sites comprising not more than 2,000 net acres available for application. Net acreage is the land available for application, excluding roads, surface water drainage, and required buffer areas. The sites comprising a single reuse project shall be contained within a ten-mile radius of a given location. There is no restriction on the number of NOIs which may be filed for reuse within any geographic area. A single reuse project may be a one-time application or may be repetitive applications to the same parcel. Filing fees are annual fees. Projects will be billed for an annual fee equaling the filing fee until the project is completed and coverage under the General Order has been terminated.
18. As a condition for the review of each individual NOI submitted for a proposed biosolids application project under the GO, the RWQCB staff responsible for issuing the NOA will:
 - a. evaluate whether the proposed discharge will occur within an area designated as having existing nitrate contamination problems and
 - b. evaluate whether the proposed discharge will pose an imminent threat of contributing to or causing exceedances of water quality standards for nitrate.
19. As a result of the review discussed in Finding No.17 , if the responsible RWQCB staff finds that either condition exists, the RWQCB staff will minimize the potential water quality impacts of the project by requiring the applicant to modify the proposed discharge activities or provide additional information to verify that the proposed discharge will not cause or contribute to violations of water quality standards. Verification that the proposed project will not cause or contribute to water quality degradation will require that sufficient information be submitted by a qualified civil engineer, agricultural engineer, professional hydrogeologist or other

qualified professional such that the RWQCB staff could make a finding that the proposed discharge will be in compliance with provisions of the GO. If the RWQCB staff finds that modifications to the proposed discharge are necessary for compliance with provisions of the GO, such modifications will consider, but will not be limited to, the following:

- a. requirements for the discharger to use the services of a certified agronomist, crop advisor, or agricultural engineer to develop additional management practices related to: 1) determining the agronomic rate for biosolids application projects that include all sources of nitrogen applied to the application site; 2) developing overall farm water, cropping, and fertility management practices; and 3) evaluating the potential for nitrate leaching or impairment of offsite groundwater use;
 - b. requirements of the discharger to provide additional groundwater monitoring in areas where groundwater is found at depths greater than 25 feet or there exist other identified local hydrogeologic conditions that could make the groundwater susceptible to contamination;
 - c. requirements of the discharger to identify whether the proposed biosolids application site is within an area where Drinking Water Source Water Assessment and Protection (DWSWAP) Program setback requirements are implemented for municipal and domestic wells; and
 - d. requirements of the discharger to consider the unique local site and hydrogeologic conditions in the design of the project and/or other groundwater quality management or regulatory programs that are currently active in the area.
20. This General Order sets minimum standards for the use of biosolids as agricultural, horticultural, silvicultural, or reclamation site soil amendments, and it does not preempt or supersede the authority of local agencies to prohibit, restrict, or control the use of biosolids subject to their control, as allowed under current law. It is the responsibility of the discharger to make inquiry and to obtain any local governmental agency permits or authorizations prior to the application of biosolids at each site.
21. Some areas in California have been designated as unique and valuable public resources. Such areas have been defined in the State law and the CCR as jurisdictional waters or preserves or have been addressed through acts specifically intended to preserve and manage the resource. This General Order is not applicable to those areas as described below:
- a. The Lake Tahoe Basin.
 - b. The Santa Monica Mountains Zone as defined by section 33105 of the Government Code.
 - c. The California Coastal Zone, as defined in and mapped pursuant to Public Resources Code (PRC) section 30103.

- d. An area within one quarter mile of a wild and scenic river, as defined by PRC section 5093.5.
- e. The Sacramento-San Joaquin Delta, as defined in CWC section 12220.
- f. The Suisun Marsh, as defined in (PRC) section 29101.
- g. The jurisdiction of the San Francisco Bay Conservation and Development Commission, as defined in Government Code section 66610.
- h. The following prohibition areas contained in the Water Quality Control Plan¹ of the Lahontan RWQCB:
 - (1) Glenshire and Devonshire Subdivisions, Town of Truckee
 - (2) Areas southwest of Piute Creek and north of Susan River and included in Sections 21, 25, 26, 27, 28, 33, 34, 35, and 36, T30N, R11E, MDB&M
 - (3) Eagle Lake Basin-Spaulding Tract, Stones-Bengard Subdivision, and Eagle's Nest Summer Home Tract
 - (4) Mono-Owens Planning Area
 - (a) Rush Creek Watershed above the outlet of Grant Lake
 - (b) Mammoth Creek Watershed, including the drainage area of the community of Mammoth Lake, and the Sherwin Creek Watershed upstream of the confluence of Sherwin and Mammoth Creeks
 - (c) Inyo County Service Area No. 1
 - i. Assessment District No. 1
 - ii. Assessment District No. 2
 - iii. Rocking K Subdivision
 - iv. City of Bishop
 - (5) Antelope Valley Planning Area
 - (a) The Antelope Hydrologic Unit above an elevation of 3,500 feet
 - (6) Mojave River Planning Area
 - (a) The Silverwood Lake Watershed
 - (b) The Deep Creek Watershed above an elevation of 3,200 feet
 - (c) The Grass Valley Creek Watershed above an elevation of 3,200 feet

¹ A detailed description of the prohibition areas can be found in the Lahontan RWQCB's Water Quality Control Plan (Basin Plan)

(d) Area north of State Highway 18 within the area commonly known as Apple Valley and Desert Knolls

(7) Hilton Creek/Crowley Lake communities

22. The biosolids applied to land under this General Order are non-hazardous decomposable wastes applied as a soil amendment pursuant to best management practices and, as such, are exempt from the requirements of Title 23, CCR, Section 2510, et seq., (Chapter 15), in accordance with Section 2511(f).
23. The construction and use of biosolids storage facilities allowed by this General Order are for short-term storage of biosolids in the event that biosolids cannot be immediately applied to the ground surface because of an unanticipated event, such as mechanical breakdown of equipment or an unseasonable rainstorm. Because of the short period of storage allowed by this General Order, the stockpiled biosolids are not a threat to the quality of underlying ground water; thus, the storage basins need not be regulated as either a waste pile or surface impoundment under Title 27 of the CCR. If long-term storage is proposed, the discharger will need to apply for a separate WDR for the long-term biosolids storage facility. Biosolids application to land associated with a project using a permitted long-term biosolids storage basin may be conducted under this General Order, if appropriate.
24. Ground water and surface waters of California have been evaluated for their maximum potential beneficial uses. Those use categories are discussed below:
 - a. The designated beneficial uses of surface waters within the State are:
 - (1) Municipal Supply (MUN)
 - (2) Agricultural Supply (AGR)
 - (3) Aquaculture (AQUA)
 - (4) Fresh Water Replenishment of Salton Sea (FRSH)
 - (5) Industrial Service Supply (IND)
 - (6) Ground Water Recharge (GWR)
 - (7) Water Contact Recreation (REC I)
 - (8) Noncontact Water Recreation (REC II)
 - (9) Warm Water Habitat (WARM)
 - (10) Cold Freshwater Habitat (COLD)
 - (11) Wildlife Habitat (WILD)
 - (12) Hydropower Generation (POW)
 - (13) Preservation of Rare, Endangered, or Threatened Species (RARE)
 - b. The designated beneficial uses of ground waters in California are:
 - (1) MUN
 - (2) IND
 - (3) AGR
 - (4) AQUA
 - (5) WILD

Some ground water and surface waters have fewer beneficial uses. Beneficial uses for specific water bodies can be found in the applicable RWQCB's Water Quality Control Plan (Basin Plan).

25. On July 22, 2004, in accordance with CEQA (PRC, Section 21000, et seq.), the SWRCB adopted a Mitigated Environmental Impact Report No. 99062108 for these General WDRs.
26. The SWRCB has notified all known interested agencies and persons of its intent to prescribe General WDRs for the reuse of biosolids as a soil amendment and has provided them with an opportunity for a public hearing and an opportunity to submit comments.
27. The SWRCB, in public meetings on March 2 and July 7, 2004, heard and considered all comments pertaining to the General Order.
28. Amendments to this General Order have been evaluated by the SWRCB in light of the Environmental Impact Report just certified and the substantial evidence before the Board, and the SWRCB finds such amendments to be consistent with the analysis contained therein. The SWRCB finds that there will be no additional potentially significant environmental impacts or substantial increase in the severity of previously disclosed environmental impacts caused by the amendments to the General Order.

IT IS HEREBY ORDERED that all dischargers that file an NOI indicating their intention to be regulated under provisions of this General Order, and all heirs, successors, or designees, in order to meet the provisions contained in Division 7 of CWC and regulations adopted thereunder, shall comply with the following:

A. PROHIBITIONS

1. The discharge of biosolids is prohibited unless the discharger has submitted an NOI, filing fee, and a pre-application report and in response to these submittals, the RWQCB has issued a Notice of Applicability, individual WDRs, or a waiver of WDRs for the discharge.
2. Applications of biosolids shall be confined to the designated use areas stated and shown in the NOI and pre-application report.
3. The discharge shall not cause or threaten to cause pollution, as defined in CWC section 13050.
4. The application of any material that results in a violation of the Safe Drinking Water and Toxic Enforcement Act (Health and Safety Code section 25249.5) is prohibited.
5. The storage, transport, or application of biosolids shall not cause a nuisance, as defined in CWC section 13050.
6. There shall be no discharge of biosolids from the storage or application areas to adjacent land areas not regulated by this General Order, to surface waters, or to surface water drainage courses.

7. From the permitted site, irrigation water runoff is prohibited for 30 days after application of biosolids if vegetation in the application area and along the path of runoff does not provide 33 feet of unmowed grass or similar vegetation to prevent the movement of biosolids from the application site.
8. Application of biosolids at rates in excess of the nitrogen requirements of the vegetation or at rates that would degrade ground water is prohibited except as allowed by Prohibition A.9.
9. Application of biosolids at rates in excess of the nitrogen requirements of the vegetation may be allowed for soil reclamation projects (as defined by land reclamation on page 4) as part of an overall plan for reclamation of sites (such as abandoned mine tailings and gravel quarries), provided the discharger can demonstrate that the application of excess nitrogen will not result in unacceptable degradation of underlying ground waters. A report prepared by a Certified Agronomist, Certified Soil Scientist, Registered Agricultural Engineer, or Registered Civil Engineer providing this demonstration shall be submitted to and approved by the RWQCB Executive Officer prior to the application of biosolids to reclamation sites at greater than agronomic rates.
10. The discharge of biosolids except as allowed for authorized storage, processing, and application sites is prohibited.
11. The application of "hazardous waste," as defined in Chapter 11, Division 4.5, Title 22 of the CCR, is prohibited.
12. Discharge of biosolids with pollutant concentrations greater than those shown below is prohibited.

<u>Constituent</u>	<u>Ceiling Concentration</u> <u>mg/kg dry weight</u>
Arsenic	75
Cadmium	85
Copper	4300
Lead	840
Mercury	57
Molybdenum	75
Nickel	420
Selenium	100
Zinc	7,500

13. The application of biosolids to water-saturated or frozen ground or during periods of precipitation that induces runoff from the permitted site is prohibited.
14. The application of Class B biosolids containing a moisture content of less than 50 percent is prohibited.

15. The application of biosolids in areas where biosolids are subject to gully erosion or washout off site is prohibited.
16. The application of biosolids to slopes exceeding 25 percent is prohibited.

B. DISCHARGE SPECIFICATIONS

1. All biosolids subject to this General Order shall comply with the applicable pathogen reduction standards listed in 40 CFR Part 503.32. In addition to those standards, all biosolids meeting Class A standards shall not have a maximum fecal coliform concentration greater than 1,000 most probable number (MPN) per gram of biosolids; or the density of salmonella, sp.² shall not be greater than three MPN per four grams.
2. All biosolids subject to this order shall comply with one of the applicable vector attraction reduction requirements specified in 40 CFR Part 503.33.
3. Biosolids application rates shall not exceed the agronomic rate for nitrogen for the crop being planted except as allowed by Prohibition No. 9 or for biosolids research projects.
4. Biosolids less than 75% moisture shall not be applied during periods when the surface wind speed exceeds 25 miles per hour as determined by the nearest calibrated regional weather station (e.g., airport, CIMS).
5. Biosolids shall not be applied in amounts exceeding the Risk Assessment Acceptable Soil Concentration as described below:

$$BC = RP - 1.8(BS)$$

Where: BC= Background Cumulative Adjusted Loading Rate
(Lbs./Acre)
 RP = 40 CFR Part 503 Cumulative Pollutant Loading Rate
(Lbs./Acre)
 BS = Actual Site Background Site Soil Concentration (mg/Kg)

And Where the Values for RP on a pollutant specific basis are given below:

Pollutant	Cumulative Pollutant Loading Rate (RP) (Lbs./Acre)
Arsenic	36
Cadmium	34
Copper	1336
Lead	267

² As determined by a USEPA approved method other than a method listed in "Standard Methods for the Examination of Water and Wastewater" 18th Edition, 1992, American Public Health Association, 1015 15th Street, NW., Washington, DC 20005; and other than the method found in Kenner, B. A. and H. P. Clark, "Detection and Enumeration of Salmonella and Pseudomonas aeruginosa," Journal of Water Pollution Control Federation, Vol. 46, No. 9, September 1974, pp. 2163-2171. Water Environment Federation, 601 Wythe Street, Alexandria, VA 22314.

Mercury	15
Molybdenum ³	16
Nickel	374
Selenium	89
Zinc	2,494

6. If biosolids are applied to a site where the soil will be tilled, biosolids shall be incorporated within 24 hours after application in arid areas and in non-arid areas during the time period beginning May 1 and ending October 31 and within 48 hours in non-arid areas during the remaining time period.
7. Grazing of domesticated animals at sites where biosolids applications have occurred will be restricted until the necessary waiting period has elapsed. Such grazing shall be deferred for at least 60 days after application of biosolids in areas with average daily (daytime) air temperatures exceeding 50°F or be deferred for at least 90 days after land application where such conditions are not met.
8. If biosolids are applied to ground surfaces having a slope greater than ten percent (10%) or if required by the RWQCB Executive Officer, a report, including an erosion control plan, shall be prepared by a Certified Soil Scientist, Certified Agronomist, Registered Agricultural Engineer, Registered Civil Engineer, or a Certified Professional Erosion and Sediment Control Specialist and submitted to the RWQCB for approval with the NOI. This report shall describe the site conditions that justify application of biosolids to the steeper slopes and shall specify the application and management practices necessary (a) to assure containment of the biosolids on the application site and (b) to prevent soil erosion. The discharger shall comply with any approved erosion control plan submitted to the RWQCB.
9. Structures conveying tail water shall be designed and maintained to minimize any field erosion. Tail water structures shall be boarded and wrapped with plastic prior to any biosolids application but removed after biosolids incorporation into the soil.
10. Biosolids distinguished as “Class B” in 40 CFR Part 503 must comply with the following:
 - a. The discharge of tail water or field runoff is prohibited within 30 days after application of biosolids for areas where biosolids have not been incorporated into the soil and where there is not a minimum of 33 feet⁴ of unmowed grass or similar vegetation bordering the application area and along the path of runoff to prevent movement of biosolids particles from the application site.
 - b. After an application of biosolids in any field, the discharger shall ensure the following:

³ Currently the USEPA has not established a value for the limitation of molybdenum. Should the USEPA establish such a cumulative pollutant limitation in 40 CFR Part 503, that limit will preempt the limit specified for molybdenum.

⁴ For sites where the topography slopes are greater than 10 percent, the minimum width of vegetative border shall be proposed in accordance to Discharge Specification No. 8 above.

- (1) For at least 30 days:
 - (a) Food, feed, and fiber crops are not harvested.
- (2) For at least 60 days after application of biosolids in areas with average daily (daytime) air temperatures exceeding 50°F or for at least 90 days after land application where such conditions are not met:
 - (a) Domesticated Animals are not grazed.
- (3) For at least 12 months:
 - (a) Public access to the site is restricted for sites with a high potential for public exposure;
 - (b) Turf is not to be harvested if the harvested turf is placed on land with a high potential for contact by the public as defined in 40 CFR Part 503.11; and
 - (c) Grazing of milking animals used for producing unpasteurized milk for human consumption is prevented if the field is used as pasture.
- (4) For at least 14 months:

Food crops with harvested parts that touch the biosolids/soil mixture and are totally above the land surface are not harvested.
- (5) For at least 20 months:

Food crops with harvested parts below the land surface are not harvested when the biosolids remain exposed on the surface for four months or longer prior to incorporation.
- (6) For at least 38 months:

Food crops with harvested parts below the land surface are not harvested when the biosolids remained exposed on the ground surface for less than four months prior to incorporation into the soil.

11. Staging and biosolids application areas shall be at least:

- a. 10 feet from property lines⁵,
- b. 500 feet from domestic water supply wells⁶,
- c. 100 feet from non-domestic water supply wells⁷,

⁵ This requirement may be waived when property lines are adjacent to properties also using biosolids as a soil amendment.

⁶ A lesser setback distance from domestic water supply wells (not to be less than 100 feet) may be used if the discharger can demonstrate to the Executive Officer that the ground water, geologic, topographic, and well construction conditions at the specific site are adequate to protect the health of individuals using the supply well.

⁷ A lesser setback distance (not to be less than 25 feet) may be used if the discharger can demonstrate to the RWQCB Executive Officer that the ground water, geologic, topographic, and well construction conditions at the specific site are adequate to protect the ground water. Not including agricultural drains.

- d. 50 feet from public roads and occupied onsite residences,
 - e. 100 feet from surface waters, including wetlands, creeks, ponds, lakes, underground aqueducts, and marshes,
 - f. 33 feet from primary agricultural drainage ways,
 - g. 500 feet from occupied non-agricultural buildings and off-site residences⁸,
 - h. 400 feet from a domestic water supply reservoir,
 - i. 200 feet from a primary tributary to a domestic water supply,
 - j. 2,500 feet from any domestic surface water supply intake, and
 - k. 500 feet from enclosed water bodies that could be occupied by pupfish.
12. Operators that produce land applied biosolids are to follow the recommendations contained in ISCORS's November 2003 draft report entitled "Assessment of Radioactivity in Sewage Sludge: Recommendations on Management of Radioactive Materials in Sewage Sludge and Ash in Publicly Owned Treatment Works" (ISCORS Technical Report 2003-04), for screening, identification, and consultation.

C. BIOSOLIDS STORAGE AND TRANSPORTATION SPECIFICATIONS

Biosolids shall be considered to be "stored" if they are placed on the ground or in non-mobile containers (i.e., not in a truck or trailer) at the application site or an intermediate storage location away from the generator/processing for more than 48 hours. Biosolids shall be considered to be "staged" if placed on the ground for brief periods of time solely to facilitate transfer of the biosolids between transportation and application vehicles.

1. Biosolids shall not be stored for more than seven (7) consecutive days prior to application.
2. Biosolids containing free liquids shall not be placed on the ground prior to application on an approved site, excluding equipment cleaning operations.
3. Biosolids shall not be stored directly on the ground at any one location for more than seven (7) consecutive days.
4. Sites for the storage of Class B biosolids shall be located, designed, and maintained to restrict public access to the biosolids.
5. Biosolids storage facilities that contain biosolids between October 1 and April 30 shall be designed and maintained to prevent washout or inundation from a storm or flood with a return frequency of 100 years.
6. Biosolids placed on site for more than 24 hours shall be covered.
7. Biosolids storage facilities shall be designed, maintained, and operated to minimize the generation of leachate and the effects of erosion.

⁸ A lesser setback from non-agricultural buildings and off-site residences (not less than 100 feet) may be allowed by the Executive Officer provided that a lesser setback is not initially opposed by the current resident within 500 feet.

8. If biosolids are to be stored at the site, a plan describing the storage program and means of complying with this General Order shall be submitted for RWQCB Executive Officer approval with the NOI. The storage plan shall also include an adverse weather plan.
9. The discharger shall operate the biosolids storage facilities in accordance with the approved biosolids storage plan.
10. The discharger shall immediately remove and relocate any biosolids stored or applied on site in violation of this General Order.
11. All biosolids shall be transported in covered vehicles capable of containing the designated load.
12. No application of Class B biosolids shall be permitted within an area defined in the General Order as having a high potential for public exposure unless the biosolids are injected into the soil.
13. All biosolids having a water content that is capable of leaching liquids shall be transported in leak proof vehicles.
14. Each biosolids transport driver shall be trained as to the nature of its load and the proper response to accidents or spill events and shall carry a copy of an approved spill response plan.
15. The discharger shall avoid the use of haul routes near residential land uses to the extent possible. If the use of haul routes near residential land uses cannot be avoided, the discharger shall limit project-related truck traffic to daylight hours.

D. PROVISIONS

1. To obtain coverage under this General Order and terminate coverage thereof, the following must take place:

- a. Coverage:

A complete NOI form and filing fee must be filed by the discharger for each proposed application site covered by these General WDRs. The NOI form may be modified by the RWQCB Executive Officer as the need arises. An NOI form is attached (Attachment A) to this General Order. Coverage does not begin until a Notice of Applicability has been issued by the applicable RWQCB's Executive Officer. No discharge shall occur until 15 days after submission of the Pre-Application Report as required in the Monitoring and Reporting Program.

- b. Coverage Termination:

- (1) A biosolids application project covered by these General WDRs may be terminated by submittal of the Final Monitoring and Reporting Program technical report and a NOT, as shown on

Attachment B of these General WDRs. The discharger(s) will be responsible for paying all annual fees for coverage under these General WDRs until approval of the NOT is granted by the RWQCB Executive Officer. For sites using Class B biosolids, termination shall not take place until 38 months after the last Class B biosolids application. The NOT form may be modified by the RWQCB Executive Officer as the need arises.

- (2) If an individual WDR Order is issued to the discharger for a project covered by this General Order, the applicability of this General Order to the discharger is automatically terminated on the effective date of the individual WDR Order.
2. Where ground water monitoring is required, as specified by the RWQCB Executive Officer or as contained in Monitoring and Reporting Program, the ground water monitoring program must be in place prior to any application of biosolids.
3. A cultural resources investigation shall be conducted before any disturbance of land that has not been disturbed previously. The cultural resources investigation will include, at a minimum, a records search for previously identified cultural resources and previously conducted cultural resources investigations of the project parcel and vicinity. This record search will include, at a minimum, contacting the appropriate information center of the California Historical Resources Information System, operated under the auspices of the California Office of Historic Preservation. In coordination with the information center or a qualified archaeologist, a determination shall be made regarding whether previously identified cultural resources will be affected by the proposed project and if previously conducted investigations were performed to satisfy the requirements of CEQA. If not, a cultural resources survey shall be conducted. The purpose of this investigation will be to identify resources before they are affected by a proposed project and avoid the impact. If the impact is unavoidable, mitigation will be determined on a case-by-case basis, as warranted.
4. The Discharger shall comply with State laws regarding disposition of Native American burials if such remains are found. If human remains of Native American origin are discovered during project activities, the discharger shall comply with State laws relating to the disposition of Native American burials, which are under the jurisdiction of the Native American Heritage Commission (Pub. Res. Code Section 5097). If human remains are discovered or recognized in any location other than a dedicated cemetery (six or more human burials at one location constitute a cemetery [Section 8100], excavation or disturbance of the site or any nearby area reasonably suspected to overlie adjacent human remains will stop until:
 - a. the county coroner has been informed of the discovery and has determined that no investigation of the cause of death is required; and
 - b. if the remains are of Native American origin,

- i. the descendants of the deceased Native Americans have made a recommendation to the landowner or the person responsible for the excavation work for means of treating or disposing of the human remains and any associated grave goods with appropriate dignity, as provided in Public Resources Code Section 5097.98, or
 - ii. the Native American Heritage Commission is unable to identify a descendant or the descendant failed to make a recommendation within 24 hours after being notified by the commission.
5. The discharger shall submit copies of each NOI to the appropriate regional office(s) of the Department of Fish and Game, local water district, City Planning Department, County Health Department(s), County Planning Department(s), and County Agricultural Commissioner(s) with jurisdiction over the proposed application site(s). Also, the discharger shall notify adjacent property owners with parcels abutting the subject land application site and, where applicable, tenants. The discharger shall submit proof to the RWQCB that all the above agencies and persons were notified. Other than compliance evaluations, the RWQCB is not responsible for the notification process. Regional Board staff will examine available records to determine if there are recorded wells at the proposed application site. No application will be permitted at the site unless the well has been properly abandoned or the set back requirements are observed.
6. The discharger shall comply with the Monitoring and Reporting Program No. 2000- which is part of this General Order and any plans required and contained within, and any revisions thereto.
7. The discharger must notify the RWQCB Executive Officer in writing at least 30 days in advance of any proposed transfer of this General Order's responsibility and coverage to a new discharger. The notice must include a new NOI for the proposed discharger, a NOT for the existing discharger, and a specific date for the transfer of this General Order's responsibility. This agreement shall include an acknowledgment that the existing discharger is liable for compliance with this General Order and for all violations up to the transfer date and that the new discharger is liable for compliance with this General Order and all violations after the transfer date.
8. Where the discharger becomes aware that it failed to submit any relevant facts in a NOI or submitted incorrect information in a NOI or in any report to the RWQCB, it shall promptly submit such facts or information.
9. The discharger shall be responsible for informing all biosolids transporters, appliers, and growers using the site of the conditions contained in this General Order.
10. The discharger must comply with all conditions of this General Order, including timely submittal of technical and monitoring reports as directed by the RWQCB Executive Officer. Violations may result in enforcement action, including RWQCB or court orders requiring corrective action or

imposing civil monetary liability or revision or rescission of the applicability of this General Order to a specific project.

11. Individuals and companies responsible for site operations retain primary responsibility for compliance with these requirements, including day-to-day operations and monitoring. Individual property owners and property managers retain primary responsibility for crop selection and any access or harvesting restrictions resulting from biosolids application. Individual owners of the real property at which the discharge will occur are ultimately responsible for ensuring compliance with these requirements. Enforcement actions for violations of this General Order may be taken against all dischargers required to comply with this General Order.
12. A copy of this General Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
13. This General Order does not convey any property rights of any sort or any exclusive privileges. The requirements prescribed herein do not authorize the commission of any act causing injury to persons or property, do not protect the discharger from his liability under federal, State, or local laws, nor do they create a vested right for the discharger to continue the waste discharge.
14. Provisions of these WDRs are severable. If any provision of these requirements is found invalid, the remainder of these requirements shall not be affected.
15. The SWRCB will review this General Order periodically and will revise requirements when necessary.
16. The discharger at all times shall properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the discharger to achieve compliance with conditions of this General Order. Proper operation and maintenance includes effective performance, adequate funding, adequate operator staffing and training, and adequate laboratory and process controls, including appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems only when necessary to achieve compliance with the conditions of this General Order.
17. The discharger shall allow the RWQCB or an authorized representative upon the presentation of credentials, valid identification with photograph, and other documents as may be required by law to:
 - a. Enter upon the discharger's premises where a regulated facility or activity is located or conducted or where records must be kept under the conditions of this General Order;
 - b. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this General Order;

- c. Inspect at reasonable times any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this General Order; and
 - d. Sample or monitor at reasonable times, any substances or parameters at any location for the purposes of assuring compliance with this General Order or as otherwise authorized by the CWC.
18. All monitoring instruments and devices used by the discharger to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary to ensure their continued accuracy. All measurement devices shall be calibrated at least once per year or more frequently to ensure continued accuracy of the devices.

Unless otherwise permitted by the RWQCB Executive Officer, all analyses shall be conducted at a laboratory certified for such analyses by the California Department of Health Services. The RWQCB Executive Officer may allow use of any uncertified laboratory under exceptional circumstances, such as when the closest laboratory to the monitoring location is outside the State boundaries and therefore is not subject to certification. All analyses shall be conducted in accordance with those methods specified in 40 CFR Part 503.8(1) through 40 CFR Part 503.8(4), 40 CFR Part 503.8(6), and 40 CFR Part 503.8(7).

19. The discharger shall report any noncompliance which may endanger human health or the environment. Any such information shall be provided orally to the RWQCB Executive Officer within 24 hours from the time the discharger becomes aware of the circumstances. A written submission shall also be provided within five days of the time the discharger becomes aware of the circumstances. The written submission shall contain (a) a description of the noncompliance and its cause; (b) the period of noncompliance, including exact dates and times; and, (c) if the noncompliance has not been corrected, the anticipated time the noncompliance is expected to continue and steps being taken or planned to reduce, eliminate, and prevent recurrence of the noncompliance with a time schedule that includes milestone dates. The RWQCB Executive Officer or an authorized representative may waive the written report on a case-by-case basis if the oral report has been received within 24 hours. Also, the discharger shall notify the Office of Emergency Services (1-800-852-7550), the State Department of Health Services, Food and Drug Branch, (916) 445-2263, and the local health department as soon as practical but within 24 hours after the incident.
20. The discharger shall retain records of all monitoring information including all calibration and maintenance records for on-site monitoring equipment (if applicable), copies of all reports required by this General Order, and records of all data used to complete the application for this General Order. Records shall be maintained for a minimum of three years from the date of the sample, measurement, report, or application. This period may be extended during the course of any unresolved litigation regarding this discharge or when requested by the RWQCB Executive Officer.

Records of monitoring information shall include:

- a. The date, exact place, and time of sampling or measurements;
- b. The individual(s) who performed the sampling or measurements;
- c. The date(s) analyses were performed;
- d. The individual(s) who performed the analyses;
- e. The analytical techniques or methods used; and
- f. The results of such analyses.

21. All application reports or information to be submitted to the RWQCB Executive Officer shall be signed and certified as follows:

- a. For a corporation--by a principal executive officer or at least the level of vice president.
- b. For a partnership or sole proprietorship--by a general partner or the proprietor, respectively.
- c. For a municipality, State, federal, or other public agency--by either a principal executive officer or ranking elected official.

22. A duly authorized representative of a person designated in Provision No. 21 of this provision may sign documents if:
- a. The authorization is made in writing by a person described in Provision No. 21, above.
 - b. The authorization specifies either an individual or position having responsibility for the overall operation of the regulated facility or activity; and
 - c. The written authorization is submitted to the RWQCB Executive Officer.

Any person signing a document under these Provisions shall make the following certification:

“I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.”

CERTIFICATION

The undersigned, Clerk to the Board, does hereby certify that the foregoing is a full, true, and correct copy of an order duly and regularly adopted at a meeting of the State Water Resources Control Board held on July 22, 2004.

AYE: Arthur G. Baggett, Jr.
 Peter S. Silva
 Richard Katz
 Gary M. Carlton
 Nancy H. Sutley

NO: None.

ABSENT: None.

ABSTAIN: None.


Debbie Irvin
Clerk to the Board

SWRCB WQ Order 2004 – 0012 - DWQ

STATE WATER RESOURCES CONTROL BOARD MONITORING AND REPORTING PROGRAM GENERAL WASTE DISCHARGE REQUIREMENTS (WDRs) FOR THE DISCHARGE OF BIOSOLIDS TO LAND FOR USE IN AGRICULTURAL, SILVICULTURAL, HORTICULTURAL, AND LAND RECLAMATION ACTIVITIES

PRE-APPLICATION REPORT

As required in Provision 1.a. of the General Order, a Pre-Application Report shall be submitted for each field or distinct application area prior to the application of biosolids in accordance with the WDRs. Where biosolids are applied on a continuing basis to a single area, the Pre-Application Report may cover ongoing operations and may not need to be submitted for each load applied. A pre-application report shall be submitted 30 days prior to the date of the proposed application. The Pre-Application Report shall be signed by the owner/operator of the biosolids application operation and by the property owner. The property owner may submit written authorization to allow a representative of the property owner, such as a tenant or land management company, to sign the Pre-Application Report.

Information in the Pre-Application Report found in **bold type** is a required field to be submitted in the Pre-Application Report. Otherwise, information that was submitted in the Notice of Intent (NOI) and has not changed or will not change is not required. The following items shall be included in the Pre-Application Report and shall be submitted to the appropriate Regional Water Quality Control Board (RWQCB):

Waste Discharge Identification System No._____

This number is established at the time the initial Notice of Intent (NOI) is submitted to the RWQCB and can be obtained at the RWQCB.

1. **Site Location/Applier Information**-A separate Pre-Application Report must be completed for each different site.

Landowner:	
Address:	
Contact:	Phone:
Site Location (including address, if any):	
Nearest Cross Street(s):	
County:	Total Size of Site:
Section(s)/Township/Range/Meridian:	
Latitude (from field center):	Longitude (from field center):

Applier:	
Address:	
Contact:	Phone:

Attach a U.S. Geological Survey 7.5 Minute map or similar map (1:24000 or larger) showing the proposed application site and surrounding properties within 2,500 feet from site boundaries. The map should show:

- a. Site topography
- b. Run-on/runoff controls
- c. Storage areas
- d. Nearby surface waters, wells, residences, and public roads
- e. Application area(s) including buffer zones (setbacks)
- f. Ground water monitoring wells (if required)
- g. Elevation

2. **Biosolids Source--** The section below must be completed **for each source of biosolids**. If additional space is required, copy this section and attach.

Wastewater Treatment Plant				
Mailing Address				
City	County	State	Zip	Phone
Contact Person				

Level of Pathogen Treatment: Class A _____ Class B _____

Description of vector attraction reduction achievement:

3. Constituent Concentrations (Each Source)

Constituent	Concentration in Biosolids, mg/kg, dry weight
Arsenic	
Cadmium	
Copper	
Lead	
Mercury	
Molybdenum	
Nickel	
Selenium	
Zinc	
pH	
Salinity	
Total Solids Content	%
Total Nitrogen	
Fecal Coliform (if applicable)	MPN/gram

Ammonia Nitrogen, as N	
Total Phosphorus, as P	
Total Potassium	
SW 846¹ Method 8080 for PCB Aroclors, Aldrin/Dieldrin	
EPA Method 8270 Semi-Volatile Organics	

Date samples collected _____
 Date samples analyzed _____
 Attach copies of all lab reports.

4. Application Area Information

Subject	Value	Applicable Unit/ Type of Measure
Quantity of Biosolids to be Applied		
Land Use Zone		
Adjacent Land Use Zones		
Application Area Size		Acres
Proposed Nitrogen Loading		Lb. plant available nitrogen/acre
Residual Nitrogen from Previous Fertilizer and Biosolids Applications¹⁰		Lb. per acre
Proposed Crop, Use		
Crop Nitrogen Usage		
Nitrogen Usage Reference		
Anticipated Avg. Appl. Rate		
Avg. Annual precipitation		
Plant tissue testing for Molybdenum(Mo)¹¹		
Plant tissue testing for Copper(Cu)³		
Plant tissue testing for Selenium(Se)³		

Attach an anticipated annual time schedule for the field operations including anticipated biosolids applications windows, seeding operations, supplemental fertilization, and cultivation/harvest.

5. Ground Water Monitoring

For biosolids application operations where minimum depth to usable ground water¹² is less than 25 feet or as specified by the RWQCB Executive Officer and where special circumstances would warrant ground water monitoring, a ground

⁹ The Discharger shall use the most recent version of SW 486 methods for detecting PCB constituents and list all Aroclor concentrations with the summation of total PCBs.

¹⁰ Attach a sheet showing calculations and all assumptions used for calculating residual Nitrogen from previous fertilizer and biosolids applications.

¹¹ The sample is a crop composite and only required where crops are used as animal feeds.

¹² Usable ground water: Ground water is defined as having either an agricultural or domestic supply source as described in the RWQCB Basin Plan.

water monitoring program, at a minimum, shall consist of three monitoring wells (one up gradient, two down gradient) for each application area and shall be in place prior to any application of biosolids if the discharger intends to or does apply biosolids more than twice within a five-year period at any particular location. A report specifying location, construction, and development details of ground water monitoring wells shall be submitted to the RWQCB for approval by the RWQCB Executive Officer prior to the installation. In addition, a mean sea level (MSL) reference elevation shall be established for each well in order to determine water elevations. The RWQCB Executive Officer, after reviewing the information submitted, may waive this requirement if it is determined that the benefit of such monitoring is not commensurate to the level of protection.

Results shall be submitted to the RWQCB 30 days prior to any biosolids application at each site and annually thereafter. Samples shall be collected from each of the monitoring wells annually and shall be analyzed for the following parameters:

<u>Parameter</u>	<u>Units</u>
Static Water Level	feet (MSL)
Total Dissolved Solids	mg/L
Sodium	mg/L
Chloride	mg/L
Nitrate	mg/L as N
Total Nitrogen	mg/L as N
pH	pH units

Initial testing shall also include the following parameters:

Arsenic	mg/L
Cadmium	mg/L
Copper	mg/L
Lead	mg/L
Mercury	mg/L
Molybdenum	mg/L
Nickel	mg/L
Selenium	mg/L
Zinc	mg/L

6. Biosolids Storage Plan (as required by Storage and Transportation Spec. No. 8)

A biosolids storage plan must be attached (even if no *on-site* biosolids storage will be provided). The biosolids storage plan should include at a minimum:

If on-site storage will be provided:

- a. Size of biosolids storage area
- b. How frequently it will be used (emergency basis only or routine use)
- c. Leachate controls
- d. Erosion controls
- e. Run-on/runoff controls

If no on-site storage will be provided:

- a. Location of off-site storage facilities
- b. Emergency storage plans

7. Erosion Control Plan (as required by Discharge Specification No. 8)

Biosolids applied to ground surfaces having a 10 percent or greater slope requires an Erosion Control Plan. The Plan should outline conditions that justify application of biosolids to the 10 percent or greater slopes and specify the application and management practices to be used to assure containment of the biosolids on the application site.

8. Spill Response and Traffic Plan (as required by Biosolids Storage and Transportation Specification No. 14)

- a. The Spill Response Plan should include at a minimum:
 - (1) Emergency contacts and notification procedures.
 - (2) Personal protective equipment requirements.
 - (3) Response instructions for spill during biosolids transport.
 - (4) Response instructions for storage facility failure.
 - (5) Response instructions if hazardous or other unauthorized material is found.
- b. The Traffic Plan should include at a minimum:
 - (1) The proposed route for all vehicles handling biosolids.
 - (2) The anticipated maximum vehicle weight.

9. Adverse Weather and Alternative Plan

Submit an Adverse Weather and Alternative Plan that details procedures to address times when biosolids cannot be applied to the site(s) due to adverse weather or other conditions (wind, precipitation, field preparation delays, access road limitations, etc.).

10. Land Productivity

A. Changes in Soil Fertility and Salinity and Resulting Effects on Productivity

"Attach a report from a certified soil scientist or a certified agronomist which evaluates the potential effects including potential nutrient imbalances, metals phytotoxicity, and excessive salinity on land productivity. The soil scientist and/or agronomist shall make recommendations, as deemed necessary, after considering the nature of the application site soils and biosolids characterization data and the need to preserve short term and long term land productivity. Those recommendations shall be reflected in the Pre-Application report regarding the proper rate of biosolids applications, any soil management

(such as supplemental fertilizers and pH adjustment), appropriate crop, and grazing practice recommendations."

B. Erosion Hazard Rating

The discharger shall submit an erosion hazard report (derived from USDA soil survey reports¹³) which assesses the proposed application site. The assessment will use the table below to determine whether soils could be degraded or land productivity reduced.

¹³ Where a soils survey report is not available for a proposed application site, the applicant shall have a qualified soil scientist determine the erosion hazard (using NRCS guidelines), unless the slope of the site is 3% or less. Sites with slopes of 3% or less will be considered to have a slight erosion hazard.

Limitations to Land Application			
Parameter	Slight	Moderate	Severe
Cation exchange capacity ^a (average milliequivalents per 100 g, 0-20 inches depth)	>15	10-15	<10
pH ^b (average 0-20 inches depth)	>6.5	5.0 to 6.5	<5.0
Erosion hazard rating ^c	None to slight	Moderate	High to severe
^a	Cation exchange capacity limits based on professional judgment.		
^b	pH limits based on U.S. Department of Agriculture (1993).		
^c	Erosion hazard limits based on professional judgment.		

Provided that the applicant, a soil scientist, or agronomist has provided written confirmation to the RWQCB that soils will not be degraded and/or land productivity will not be reduced as a result of nutrient imbalances, metals-related phytotoxicity, or adverse salinity effects, biosolids may be applied on any site having a “slight” limitation as defined in the table. At sites having a “moderate” limitation, biosolids may be applied only where the crop is not known to be particularly sensitive to metals and nutrient imbalances or is not known to be bioaccumulative of heavy metals. Sites having a “severe” limitation are excluded from eligibility under the GO. Sites with a slope of greater than 20% shall not accept biosolids unless those sites will be immediately covered by sod or a sufficient mulch cover to control erosion.

11. A biological site assessment is required in areas where natural terrestrial habitat (previously undisturbed lands) and fallow lands (as defined in Findings No. 3m in the General Order) exist and are planned for biosolids applications. The assessment shall be conducted to identify any special-status plant and wildlife species onsite, submitted as part of the Pre-Application Report, and shall be conducted by a qualified biologist. This report must be forwarded to the appropriate regional office of the DFG and the Endangered Species Unit of the USFWS in Sacramento for review and approval of the mitigation strategy, as appropriate. If there are no special-status species present, RWQCB may continue with the project evaluation. If special-status species could be affected, the project will not be authorized under the GO unless the applicant submits a plan to mitigate for any significant impacts on special-status species, obtains the appropriate permits, and agrees to implement the mitigation.

ANNUAL REPORTING

1. Ground Water Monitoring (if required in the Pre-Application Report)

Samples shall be collected from each of the monitoring wells annually and shall be analyzed for the following parameters:

<u>Parameter</u>	<u>Units</u>
Static Water Level	feet (MSL)
Total Dissolved Solids	mg/l
Sodium	mg/l
Chloride	mg/l
Nitrate	mg/l as N
Total Nitrogen	mg/l as N
pH	pH units
Arsenic (As)	mg/l
Selenium (Se)	mg/l
Molybdenum (Cu)	mg/l

2. Application Information

Quantity of Biosolids Applied		Dry tons
Application Area Size		Acres
Total Nitrogen Concentration in Biosolids		mg/kg
Nitrogen Loading		Lb. plant avail. Nitrogen per acre
Residual Nitrogen ¹⁴		Lbs. per acre
Crop		
Amount of Crop Produced		Specify units
Plant tissue testing for Molybdenum (mo) ⁶		
Plant tissue testing for Copper (cu), ⁶		
Plant tissue testing for Selenium (Se) ¹⁵		

¹⁴ As determined by field soil nitrogen testing in an 18 inch depth.

¹⁵ Crop composite and only required where crops are used as animal feeds.

3. Pollutant Loadings for Each Application Site

Pollutant	Total Loadings from Previous Years (kg/ha)	Loading This Year (kg/ha)	Background Soils Conc. (kg/ha) (6" depth)	Cumulative Metal Load to Date (kg/ha)	Percent Cumulative Limit to Date
Arsenic					
Cadmium					
Copper					
Lead					
Mercury					
Molybdenum					
Nickel					
Selenium					
Zinc					

4. Constituent Concentrations (Each Source)

Constituent	Concentration in Biosolids, (mg/kg, dry weight)
Arsenic	
Cadmium	
Copper	
Lead	
Mercury	
Molybdenum	
Nickel	
Selenium	
Zinc	
Total Solids Content	%
Total Nitrogen	
Fecal Coliform	MPN/gram
Ammonia Nitrogen, as N	
Total Phosphorus, as P	
Total Potassium	
SW 846 ¹⁶ Method 8080 for PCB Aroclors, Aldrin/Dieldrin	
EPA Method 8270 Semi-Volatile Organics	

5. Site Map

Provide a site map identifying the area(s) of application clearly showing each field to which biosolids have been applied and crop planted.

¹⁶ The discharger shall use the most recent version of SW 486 methods for detecting PCB constituents and list all Aroclor concentrations with the summation of total PCBs.

6. **40 CFR Part 503**

Attach a copy of the generator's monitoring report for compliance with the 40 CFR Part 503.

GENERAL REPORTING

1. Pre-Application Reports shall be submitted for RWQCB staff review and approval at least 30 days prior to application of biosolids. Annual Reports covering the period between January 1 to December 31 shall be submitted by February 15 of the following year. If no applications occurred during the year, the discharger shall submit a report indicating that no discharge occurred during the year.
2. The collection, preservation and holding times of all samples shall be in accordance with U.S. Environmental Protection Agency approved procedures. A laboratory certified by the California Department of Health Services to perform the required analyses shall conduct all analyses, except soil nitrogen and plant tissue samples for selenium, copper and molybdenum. Analysis for soil nitrogen and plant tissue concentrations of selenium and molybdenum shall participate in a program similar to the North American Proficiency Testing Program (NAPT) operated by the Soil Science of America. The RWQCB Executive Officer may allow use of an uncertified laboratory in accordance with Provision 18.
3. If there is no discharge during a required reporting period, the discharger shall submit a letter report to the RWQCB indicating that there has been no activity during the required reporting period.
4. Each report shall be signed and contain the following certification:

“I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment”
5. A duly authorized representative of the discharger may sign the documents if:
 - a. The authorization is made in writing by the person described above;
 - b. The authorization specified an individual or person having responsibility for the overall operation of the regulated disposal system; and
 - c. The written authorization is submitted to the RWQCB Executive Officer.
6. The discharger shall arrange the data in tabular form so that the specified information is readily discernible. The data shall be summarized in such a manner as to clearly illustrate whether the facility is operating in compliance with waste discharge requirements.
7. Report immediately (within 24 hours) to the RWQCB Executive Officer and Director of County Environmental Health by telephone with a follow-up letter any discharge which threatens the environment or human health. During non-business hours, report to the Office of Emergency Services by telephone at 1-800-852-7550.

8. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the RWQCB.

State of California

State Water Resources Control Board

NOTICE OF INTENT

TO COMPLY WITH THE TERMS OF GENERAL PERMIT ORDER NO . 2000-___-DWQ
FOR THE DISCHARGE OF BIOSOLIDS TO LAND
FOR USE IN AGRICULTURAL, SILVICULTURAL, HORTICULTURAL AND LAND RECLAMATION ACTIVITIES

ATTACHMENT A

Mark Only One Item	1. ☐ New Discharge Under MODEL Permit 2. ☐ Change of Information-WDID # _____
--------------------	--

I. Property Owner (Required)

Name				
Mailing Address				
City	County	State	Zip	Phone
Contact Person		(check one) Owner _____ Operator _____ Owner/operator _____		

II. Generator (Required . If more than one generator, attach the information and ensure that the signature block is copied, signed and attached.)

Name				
Mailing Address				
City	County	State	Zip	Phone
Contact Person				

III. Site Operator/Property Manager (if any)

Name				
Mailing Address				
City	County	State	Zip	Phone
Contact Person				

IV. Billing Address

Name				
Mailing Address				
City	County	State	Zip	Phone
Contact Person				

STATE USE ONLY

WDID: □□□□□□□□□□	Regional Board Office: □□	Date NOI Received: _____	Date NOI Processed: _____
Fee Amount Received: \$ _____		Check #: _____	

V. Site Operator

Name				
Mailing Address				
City	County	State	Zip	Phone
Contact Person				

VI. Hauler Information

Name				
Mailing Address				
City	County	State	Zip	Phone
Contact Person				
Type of Transportation				

VII. Site Location

Street (including address, if any)	
Nearest Cross Street(s)	
County:	Total Size of Site (acres):
Township/Range/Section T _____, R _____, Section _____, _____ B&M Latitude/Longitude (From Center): _____ Deg. _____ Min. _____ Sec N. _____ Deg. _____ Min. _____ Sec. W	
Attach a map of at least 1:24000 (1" = 2000") showing the proposed application site (e.g., USGS 7.5" topographic map). The map should also show run-on/runoff controls, storage areas, nearby surface waters, wells and residences, the application areas including setback and buffer zones.	

VIII. Application Area Information

Subject	Value	Applicable Unit/ Type of Measure
Quantity of Biosolids to be Applied		dry tons per year
Total Biosolids Application Proposed		dry tons
Land Use Zone		
Adjacent Land Use Zones		
Application Area Size		acres
Proposed Nitrogen Loading		lb. Plant Available Nitrogen/acre
Proposed Crop, Use		crop type, human/animal/neither
Crop Nitrogen Usage		1b. Nitrogen/year
Nitrogen Usage Reference		
Depth of Root Zone for Crop Being Planted		inches
Will Setback Limits Be Met?		Yes or No
Distance to Nearest Inhabited Dwelling		feet/miles
Public Access Controls		Specify Type
Runoff Controls		Attach plans
Prevailing Wind Direction		
Minimum Depth to Ground Water		feet
How Minimum Depth to Ground Water is Determined		

Anticipated Average Daily Application Rate		dry tons/day
Source of Water for Crop		
Average Annual Precipitation		inches/year

Attach an anticipated annual time schedule for the field operations including anticipated biosolids applications windows, seeding operations, supplemental fertilization, and cultivation/harvest.

IX. Soil Constituent Concentrations (Each Source)

Constituent	Concentration in Soil, mg/kg, dry weight
Arsenic	
Cadmium	
Copper	
Lead	
Mercury	
Molybdenum	
Nickel	
Selenium	
Zinc	
pH	
Estimated Permeability	cm/sec
Cation Exchange Capacity	meq/100g
Total Nitrogen	
Ammonia Nitrogen, as N	
Total Phosphorus, as P	
Total Potassium	

X Have any proposed biosolids application sites been fallow for more than one year? ☐ YES ☐ NO

XI Are there existing agricultural, silvicultural, or horticultural operations at all the proposed application sites? ☐ YES ☐ NO

XII Is it known whether any locations within the proposed land application site contain biologically unique or sensitive natural communities?
☐ YES ☐ NO

If natural terrestrial habitats are present on the project site, a biological site assessment must be conducted to determine whether biologically unique or sensitive natural communities occur and whether they could be disturbed by the application of biosolids; this report must be forwarded to the appropriate regional office of DFG and the Endangered Species Unit of the USFWS in Sacramento for review and approval of the mitigation strategy, as necessary. If biologically unique or sensitive natural communities are present and more than 10% or 10 acres will be disturbed, whichever is less, the project will not be authorized under the GO unless the applicant submits a plan to mitigate for any significant impacts on biologically unique or sensitive natural communities and agrees to implement the mitigation.

XIII Biosolids Storage Plan (as required by Biosolids Storage and Transportations Spec. No. 8)

A biosolids storage plan must be attached (if no on-site biosolids storage will be provided, a contingency plan for inclement weather operation must be provided). The biosolids' storage plan should include at a minimum:

If on-site storage will be provided:

- Size of biosolids storage area
- How frequently it will be used (emergency basis only or routine use)
- Leachate controls
- Erosion controls
- Run-on/runoff controls

If no on-site storage will be provided:

- Location of off-site storage facilities
- Emergency storage plans

XIV Erosion Control Plan (if applicable) (as required by Discharge Specification No. 8)

Biosolids applied to ground surfaces having a 10 percent or greater slope requires an Erosion Control Plan. The Plan should outline conditions that justify application of biosolids to the 10 percent or greater slopes and specify the application and management practices to be used to assure containment of the biosolids on the application site.

XV. Spill Response and Traffic Plan (as required by Biosolids Storage and Transportation Spec. No. 14)

a. The Spill Response Plan should include at a minimum:

1. Emergency contacts and notification procedures
2. Require personal protective equipment requirement
3. Response instructions for spill during biosolids transport
4. Response instructions for storage facility failure
5. Response instructions if hazardous or other unauthorized material is found

b. The Traffic Plan should include at a minimum:

1. The proposed route for all vehicles handling biosolids
2. Describe the anticipated maximum vehicle weight

XVI. Adverse Weather and Alternative Plan: (as required by Biosolids Storage and Transportation Spec. No. 8)

Submit an Adverse Weather and Alternative Plan that details procedures to address times when biosolids cannot be applied to the site(s) due to adverse weather or other conditions (wind, precipitation, field preparation delays, access road limitations, etc.).

XVII. CERTIFICATION

"I certify under penalty of law that this document and all attachments were prepared under my direction and supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment." In addition, I certify that the provisions of the permit, including the criteria for eligibility, will be complied with.	
Signature of Owner/Operator of Spreading Operations	Title
Printed or Typed Name	Date
Signature of Property Owner	Title
Printed or Typed Name	Date
Signature of Site Operator/Manager (if any)	Title
Printed or Typed Name	Date

State of California
State Water Resources Control Board

NOTICE OF TERMINATION

TO COMPLY WITH THE TERMS OF GENERAL PERMIT ORDER NO. 2000-___-DWQ
FOR THE DISCHARGE OF BIOSOLIDS TO LAND
FOR USE IN AGRICULTURAL, SILVICULTURAL, HORTICULTURAL AND LAND RECLAMATION ACTIVITIES

ATTACHMENT B

WDID # _____

I. Property Owner

Name				
Mailing Address				
City	County	State	Zip	Phone
Contact Person				

II. Generator

Name				
Mailing Address				
City	County	State	Zip	Phone
Contact Person				

III. Owner/Operator of spreading operations

Name				
Mailing Address				
City	County	State	Zip	Phone
Contact Person		(check one) Owner _____ Operator _____ Owner/operator _____		

IV. Site Operator/Property Manager (if any)

Name				
Mailing Address		Contact Person		
City	County	State	Zip	Phone

V. Billing Address

Name				
Mailing Address		Contact Person		
City	County	State	Zip	Phone

VI. Hauler Information

Name				
Mailing Address				

City	County	State	Zip	Phone
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VII. Site Location

Street (including address, if any)	
Nearest Cross Street(s)	
County:	Total Size of Site (acres):
Township/Range/Section T _____, R _____, Section _____, _____ B&M	
Latitude/Longitude (From Center): _____ Deg. _____ Min. _____ Sec N. _____ Deg. _____ Min. _____ Sec. W	
Attach a map of at least 1:24000 (1" = 2000") showing the proposed application site (e.g., USGS 7.5" topographic map). The map should also show run-on/runoff controls, storage areas, nearby surface waters, wells and residences, the application areas including setback and buffer zones .	

VIII. Application Area Information

Subject	Value	Applicable Unit/ Type of Measure
Quantity of Biosolids Applied		dry tons per year
Application Area Size		acres
Nitrogen Loading		lb. Plant Available Nitrogen/acre
Crop, Use		crop type, human/animal/neither
Crop Nitrogen Usage		1b. Nitrogen/year
Nitrogen Usage Reference		
Last Date of Class B Biosolids Application		Date
Public Access Controls		Specify Type

IX. Attached is the Annual Monitoring and Reporting Report for the ☐ current ☐ year. Yes No

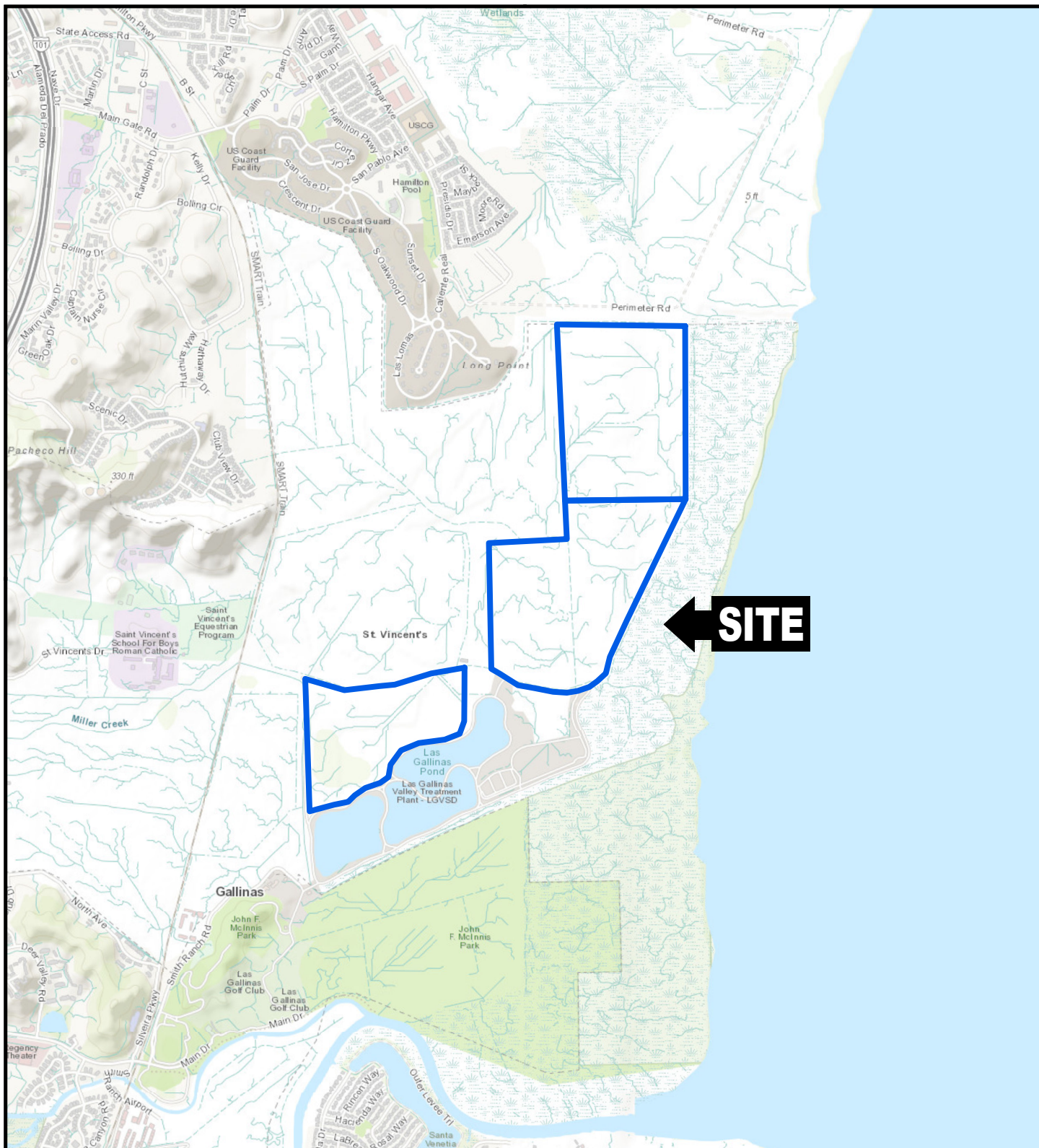
X. CERTIFICATION

"I certify under penalty of law that this document and all attachments were prepared under my direction and supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment." In addition, I certify that the provisions of the permit, including the criteria for eligibility, will be complied with.	
Signature of Generator	Title
Printed or Typed Name	Date
Signature of Property Owner	Title
Printed or Typed Name	Date
Signature of Site Operator/Manager (if any)	Title
Printed or Typed Name	Date

STATE USE ONLY

WDID: □□□□□□□□□□	Regional Board Office: □□	Date NOI Received: _____	Date NOI Processed: _____
Fee Amount Received: \$ _____		Check #: _____	

**ATTACHMENT 3: Project Location and Biosolids Application Area
Maps**



QUADRANGLE LOCATION



Title:

SITE LOCATION MAP

SAN RAFAEL, CA

Prepared for: LAS GALLINAS VALLEY SANITARY DISTRICT

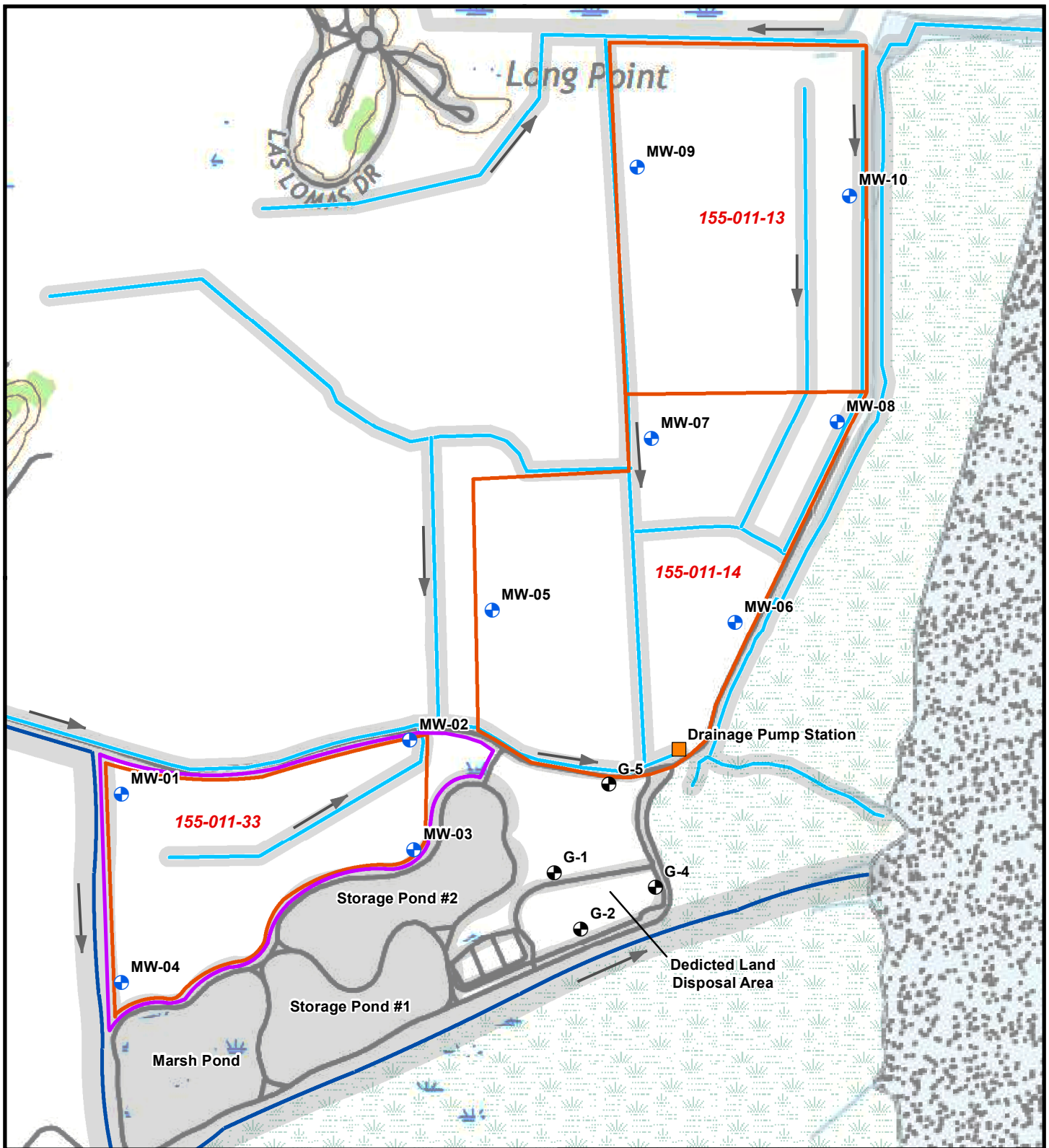
ROUX

Compiled by: S.A.	Date: 06/23/21
Prepared by: L.M.	Scale: AS SHOWN
Project Mgr: M.F.	Project: LGVSD
File: LGVSD.100.01_mf.mxd	

FIGURE

1

S:\GIS\PROJECTS\PROPOSALS\LGVS\100\LGVS\100.02.MXD



LEGEND

-  PROPOSED GROUNDWATER MONITORING WELL
-  EXISTING MONITORING WELLS
-  DRAINAGE PUMP STATION
-  MILLER CREEK
-  WATER FEATURE
-  ELEVATED ROADWAY/TRAIL
-  DRAINAGE FLOW DIRECTION
-  SITE BOUNDARY
-  BUFFER ZONE



500 0 500'

Title:

SITE PLAN

SAN RAFAEL, CA

Prepared for: LAS GALLINAS VALLEY SANITARY DISTRICT

ROUX

Compiled by: S.A.

Date: 07/27/21

Prepared by: L.M.

Scale: AS SHOWN

Project Mgr: M.F.

Project: LGVSD

File: LGVSD.100.02.mxd

FIGURE

2